Psychotic Disorders & Schizophrenia

Psychiatry, Clinical Lecture | Dr. Hussain M. Khrad (Consultant Psychiatrist, KFSHRC-Jeddah) | Source: `Psychotic-Disorders.pdf` (78 slides)

> **Legend:** ⭐ = high-yield · *[added]* = not on slides. Images marked "Slide N" come from the lecture. All 4 MCQ answers come from the lecture's own answer slides.
>
> 🔗 **Bridge from earlier lectures:**
>
> - **MSE:** delusion types, hallucination modalities, the thought-form ladder, catatonia and negative symptoms are the **vocabulary** of this lecture. "Non-auditory = organic" returns as a red flag.
> - **History Taking:** organic red flags, medical mimics, the delirium case (Mr. S), and "ask the family: is this normal for him?"
> - **Personality Disorders:** Cluster A (paranoid, schizotypal) sits **short of** psychosis; schizotypal is on the **schizophrenia spectrum**.
> - **Pituitary I (prolactin):** dopamine tonically **inhibits prolactin**, so D2 blockade → hyperprolactinaemia (tuberoinfundibular pathway).
> - **ECG/Arrhythmia:** antipsychotic **QT prolongation → torsades** (haloperidol IV, ziprasidone, pimozide).

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1. Title / Objectives

Domain	Objective	Likely tested as
Knowledge	Define **psychosis**; signs and symptoms	Symptom vs diagnosis; 6 domains
Knowledge	Classify **delusions and hallucinations**	Bizarre vs non-bizarre; first-rank; modality → cause
Knowledge	⭐ Psychotic disorders and **the duration that separates them**	< 1 month / 1–6 months / > 6 months; 2-week schizoaffective rule
Knowledge	⭐ **DSM-5-TR criteria** for schizophrenia	Criterion A: ≥ 2, ≥ 1 from the first 3
Knowledge	Aetiological hypotheses	Dopamine pathways; NMDA; neurodevelopmental
Knowledge	Course, prognosis, treatment	15/30/55; relapse 60–70% at 1 y off meds
Skill	⭐ **Psychosis vs delirium** at the bedside	Post-op 72-year-old
Skill	Medical and drug causes to exclude	Anti-NMDA, steroids, anticholinergics
Skill	⭐ **Extrapyramidal and metabolic harms**	Dystonia → akathisia → parkinsonism → TD; NMS; clozapine
Values	Medical illness, not a moral/spiritual failing; **culture vs delusion**; involve the family; physical health; non-stigmatising language	Evil eye / ruqyah case

Why it matters (slide 3): whatever your specialty, you will **prescribe drugs that cause psychosis** and be asked to see confused patients on your own ward.

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2. Main Content (Lecture Order)

2.1 Opening Case: Fahad, 21 (Slide 5)

Presentation:

- engineering student, brought by his father
- **8 months of withdrawal**: stopped meeting friends, grades collapsed, stays in his room
- **2 months:** convinced a **neighbour placed a monitoring device** in the flat; taped over the vent
- hears **two male voices commenting** on what he does, sometimes **ordering him to stay silent**
- **not confused**, oriented
- **no substances, no medication, no medical history**
- **uncle "had the same illness" in his twenties**

⭐ **What is NOT here:** no clouding of consciousness, no fluctuation, no substance, no medical illness, and **8 months of declining function**. *[added]* This is the picture of schizophrenia:

- persecutory delusion
- 3rd-person commentary and command hallucinations
- negative prodrome
- duration > 6 months
- family history

2.2 Psychosis vs Psychotic Disorder (Slide 6) ⭐

Psychosis: the SYMPTOM	Psychotic disorders: the DIAGNOSES
A gross impairment in **REALITY TESTING**	A group of disorders in which psychosis **DOMINATES**
Inability to distinguish real from unreal	Brief psychotic · schizophreniform · schizophrenia
⭐ **A symptom, not a disease**, like fever or breathlessness	Schizoaffective · delusional disorder
Caused by **anything that disturbs brain function**	Psychosis also appears in **mood disorders, delirium, dementia, substance use**

⭐ **"Psychotic" on a referral tells you nothing about the diagnosis.** It is the starting point of the work-up.

2.3 When Psychosis-Like Experience Is Normal (Slide 7)

Situation	Features
Hypnagogic / hypnopompic	Vivid perceptions on **falling asleep / waking**. Common, brief, confined to the sleep–wake transition. Also in **narcolepsy**
Bereavement	Hearing or briefly seeing the deceased in the weeks after a death: a **culturally normal grief phenomenon**. The person knows it isn't real; function preserved
Severe sleep deprivation	Misperceptions → frank hallucinations in healthy people. **Fully reverses with sleep** (residents on long shifts)

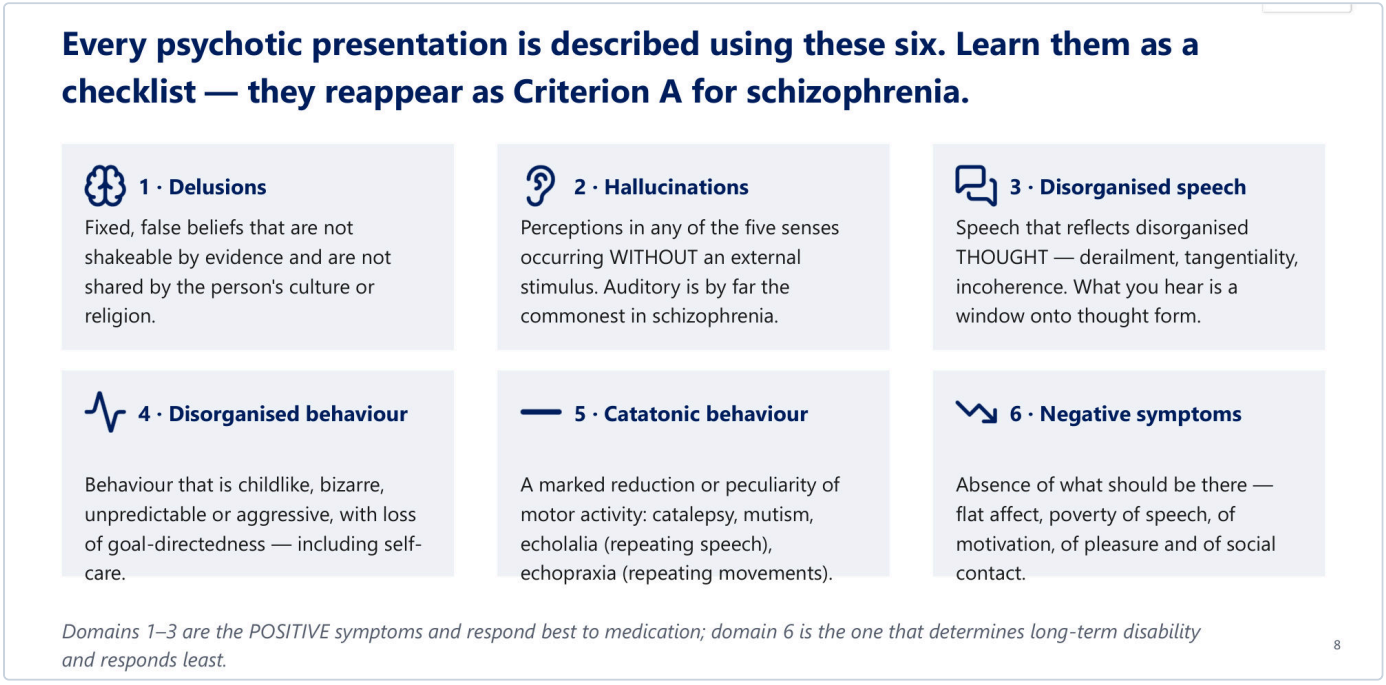
⭐ **Discriminators:**

1. **insight preserved**
2. **brief and situation-bound**
3. **no functional decline**

2.4 The Six Domains of Psychosis (Slide 8) ⭐

#	Domain	Definition
1	**Delusions**	Fixed, false beliefs, not shakeable by evidence, **not shared by culture/religion**
2	**Hallucinations**	Perceptions **without an external stimulus**; **auditory** is by far the commonest in schizophrenia
3	**Disorganised speech**	Reflects disorganised **thought**: derailment, tangentiality, incoherence
4	**Disorganised behaviour**	Childlike, bizarre, unpredictable, aggressive; loss of goal-direction, including **self-care**
5	**Catatonic behaviour**	**Catalepsy, mutism, echolalia, echopraxia**
6	**Negative symptoms**	Absence of what should be there: flat affect, poverty of speech, motivation, pleasure, social contact

- ⭐ **Domains 1–3 = POSITIVE**; they **respond best** to medication.
- **Domain 6** determines **long-term disability** and responds **least**.
- These 6 reappear as **Criterion A** (catatonic and disorganised behaviour combined into one item).



Slide 8: The six domains of psychosis

2.5 Delusions (Slides 9–14) ⭐⭐

Definition (slide 9):

- a **FIXED, FALSE** belief, held with **absolute conviction** despite contradictory evidence
- **NOT accepted by the person's culture, religion or subculture**
- not amenable to reasoning
- often (not always) **acted upon**

Bizarre vs non-bizarre	Not a delusion
BIZARRE: clearly implausible ("a **chip implanted** in my brain")	**OVERVALUED IDEA**: unreasonable but held **with some doubt**
NON-BIZARRE: possible but false ("my **wife is having an affair**")	**OBSESSION**: **resisted** and recognised as **one's own**

Bizarre vs non-bizarre	Not a delusion
⭐ **Non-bizarre delusions = the hallmark of DELUSIONAL DISORDER**	—

Types by theme (slides 10–11):

Type	Belief	Classically seen in / note
⭐ **Persecutory (paranoid)**	Followed, watched, plotted against, harmed	**The commonest**; schizophrenia, delusional disorder
Grandiose	Special, chosen, immensely wealthy, special powers	**Mania**; schizophrenia
Nihilistic	Dead or dying; world/organs no longer exist	⭐ **Psychotic depression, esp. elderly (Cotard)**
Somatic	A body part/function missing or diseased with no evidence ("my kidney was removed")	Psychotic depression; schizophrenia; delusional disorder (somatic)
Religious	Special relationship with God, usually with grandiosity	Schizophrenia; mania. **Judge against the patient's own community**
Erotomanic	A higher-status person/public figure loves the patient	**De Clérambault**; may lead to **stalking**
Jealous	Partner unfaithful, on no evidence	**Othello**; ⭐ associated with **alcohol misuse and violence risk**
Of reference	External cues aimed at the patient ("the presenter is talking to me")	**Very common; often the earliest symptom** families notice
Thought broadcasting	Thoughts can be heard by others	**Passivity**; Schneiderian first-rank
Thought insertion/withdrawal	Thoughts put into/removed from the mind by an outside force	**Passivity**: the boundary of the self is breached
Delusion of control	Actions, feelings, impulses directed by an external agent	"**Made acts / made feelings**"

- A scar-free abdomen and normal ultrasound **don't shift** a somatic delusion. That **unshakeability** separates it from **health anxiety**.
- ⭐ **Passivity phenomena:** the patient experiences their own mental life as coming from **outside**. **Ask explicitly**; patients rarely volunteer them.

Belief, Culture and Religion (Slides 12–14) ⭐

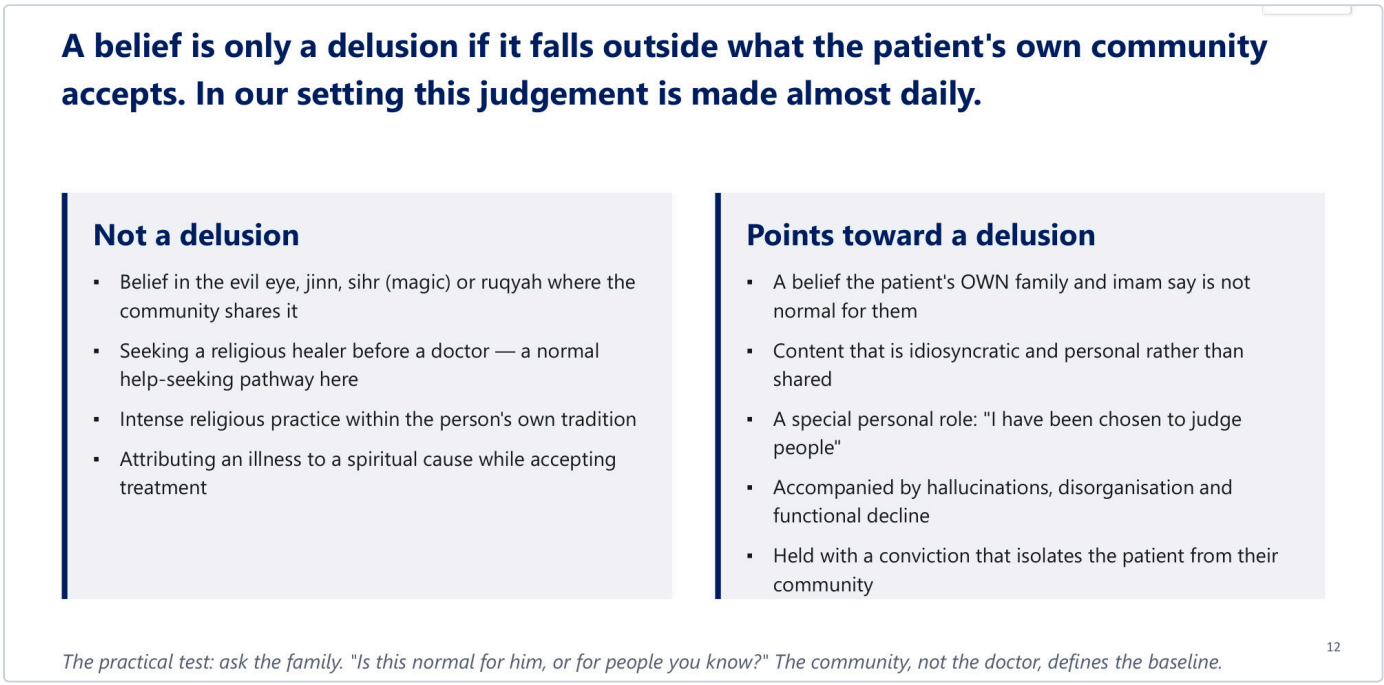
✅ Not a delusion	⚠️ Points toward a delusion
Belief in **evil eye, jinn, sihr (magic), ruqyah** where the community shares it	A belief the patient's **OWN family and imam** say is not normal for them
Seeking a **religious healer before a doctor**: a normal help-seeking pathway here	Content that is **idiosyncratic and personal** rather than shared
Intense religious practice within the person's tradition	A **special personal role** ("I have been chosen to judge people")
Attributing illness to a spiritual cause **while accepting treatment**	Accompanied by **hallucinations, disorganisation, functional decline**
—	Held with a conviction that **isolates** the patient from their community

⭐ **Practical test: ask the family, "Is this normal for him, or for people you know?"** The community, not the doctor, defines the baseline.

Case (slides 13–14):

- 34-year-old man believes the family is affected by the **evil eye**; requests ruqyah
- works normally, prays with family, sleeps well
- **whole extended family shares the belief**; no hallucinations, no decline
- referral: "delusional beliefs, assess for antipsychotics"

→ ⭐ **B: a culturally shared belief, not a delusion.** Treating him would expose him to metabolic/EPS risk for no benefit and lose the family's trust. **The answer would change with:** an idiosyncratic version the family find strange, added hallucinations, or functional decline.



Slide 12: Belief, culture and religion

2.6 Perception (Slides 15–16) ⭐

Hallucination	Illusion	Pseudohallucination
NO external stimulus	**MISPERCEPTION of a real** stimulus	Perceived in **INNER** space, "inside my head"
Real, located in **external space**	A dressing-gown on a door seen as a person	**Insight usually retained**
Any of the 5 modalities	Common in **delirium, poor light, high anxiety**	Seen in **dissociation, personality disorder, grief**
Insight typically absent	—	**Not by itself evidence of psychotic disorder**

Key question: ⭐ "Where does the voice come from?" This separates a hallucination from a pseudohallucination.

Modality	Patient describes	Think
⭐ **Auditory**	Voices, commonly **2+ talking about** the patient, or **commanding**	**Commonest in schizophrenia.** Command → ask about **compliance**

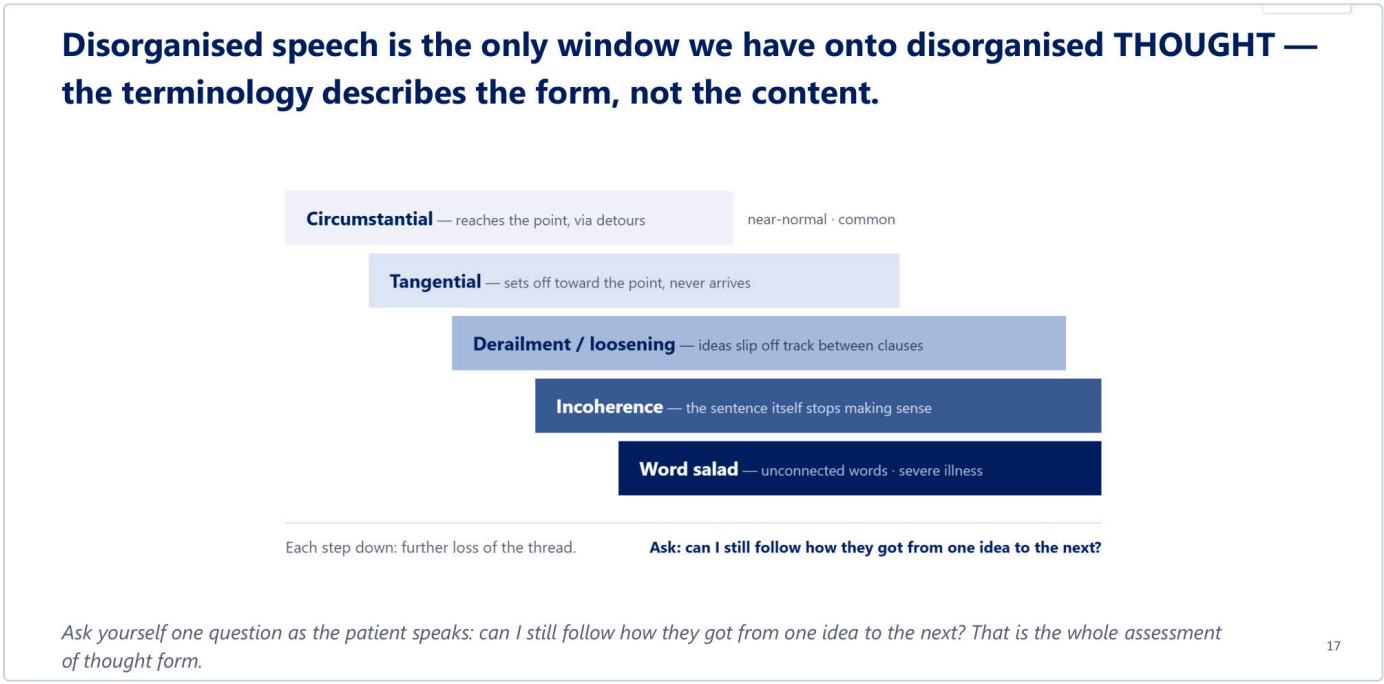
Modality	Patient describes	Think
Visual	People, animals, figures	⭐ **DELIRIUM, dementia (Lewy body), substances, epilepsy**
Olfactory	Unpleasant smell (often **burning**) preceding other symptoms	⭐ **Temporal lobe epilepsy aura**: investigate
Tactile	Insects crawling on/under the skin (**formication**)	**Stimulant intoxication (cocaine, amphetamine); alcohol withdrawal**
Gustatory	Persistent strange/**metallic** taste	Rare: epilepsy, drug effects

⭐ **The rule that saves lives: a NON-AUDITORY hallucination in a first presentation is ORGANIC until proven otherwise.**

2.7 Thought Form: the Ladder of Severity (Slide 17)

Circumstantial (reaches the point via detours; near-normal) → **Tangential** (never arrives) → **Derailment/loosening** (ideas slip between clauses) → **Incoherence** (the sentence stops making sense) → **Word salad** (unconnected words; severe illness).

Ask: ⭐ **"Can I still follow how they got from one idea to the next?"** That is the whole assessment of thought form.



Slide 17: Thought-form ladder

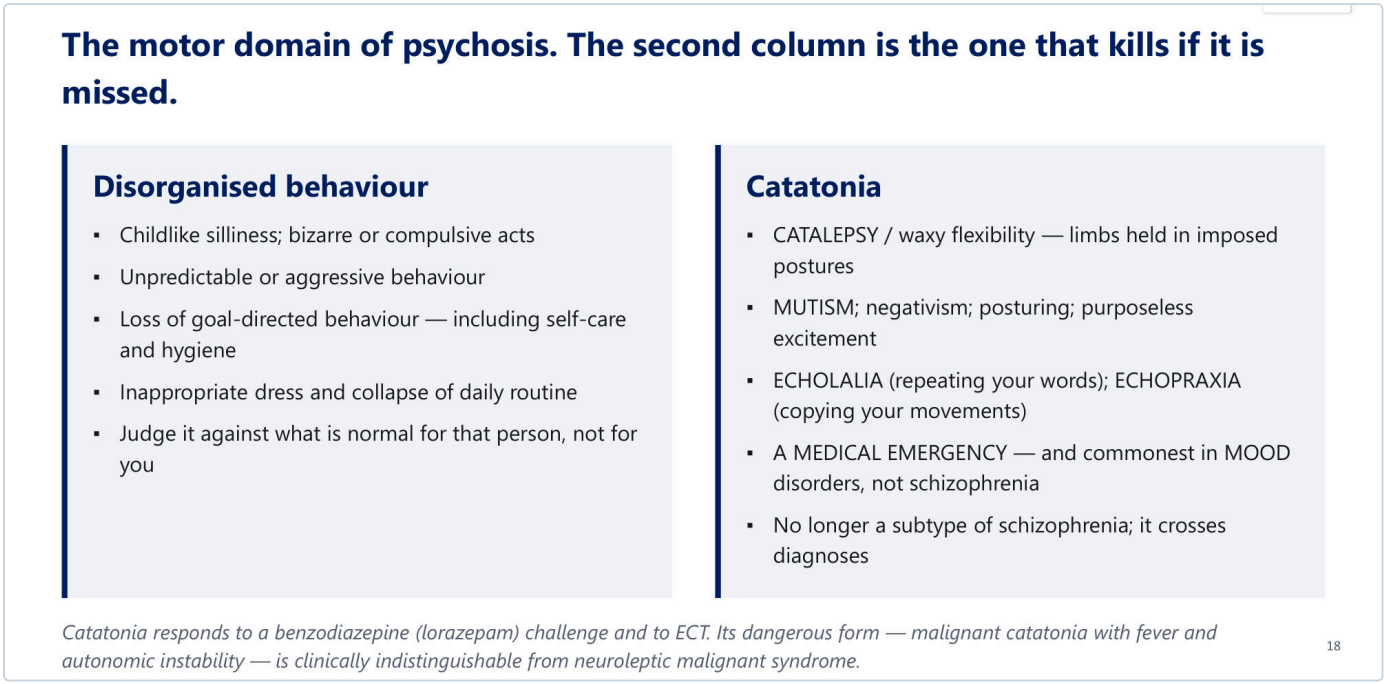
2.8 Disorganised Behaviour and Catatonia (Slide 18) ⭐

Disorganised behaviour	⭐ Catatonia
Childlike silliness; bizarre or compulsive acts	**CATALEPSY / waxy flexibility**: limbs held in imposed postures
Unpredictable or aggressive	**MUTISM; negativism; posturing; purposeless excitement**
Loss of goal-directed behaviour, including self-care	**ECHOLALIA** (repeats your words); **ECHOPRAXIA** (copies your movements)
Inappropriate dress; collapse of routine	⭐ **A MEDICAL EMERGENCY**, and **commonest in MOOD disorders, not schizophrenia**

Disorganised behaviour	⭐ Catatonia
Judge against what is normal for **that person**	**No longer a subtype of schizophrenia**; it crosses diagnoses

Catatonia:

- responds to a ⭐ **lorazepam challenge** and to **ECT**
- **malignant catatonia** (fever + autonomic instability) is clinically **indistinguishable from NMS**

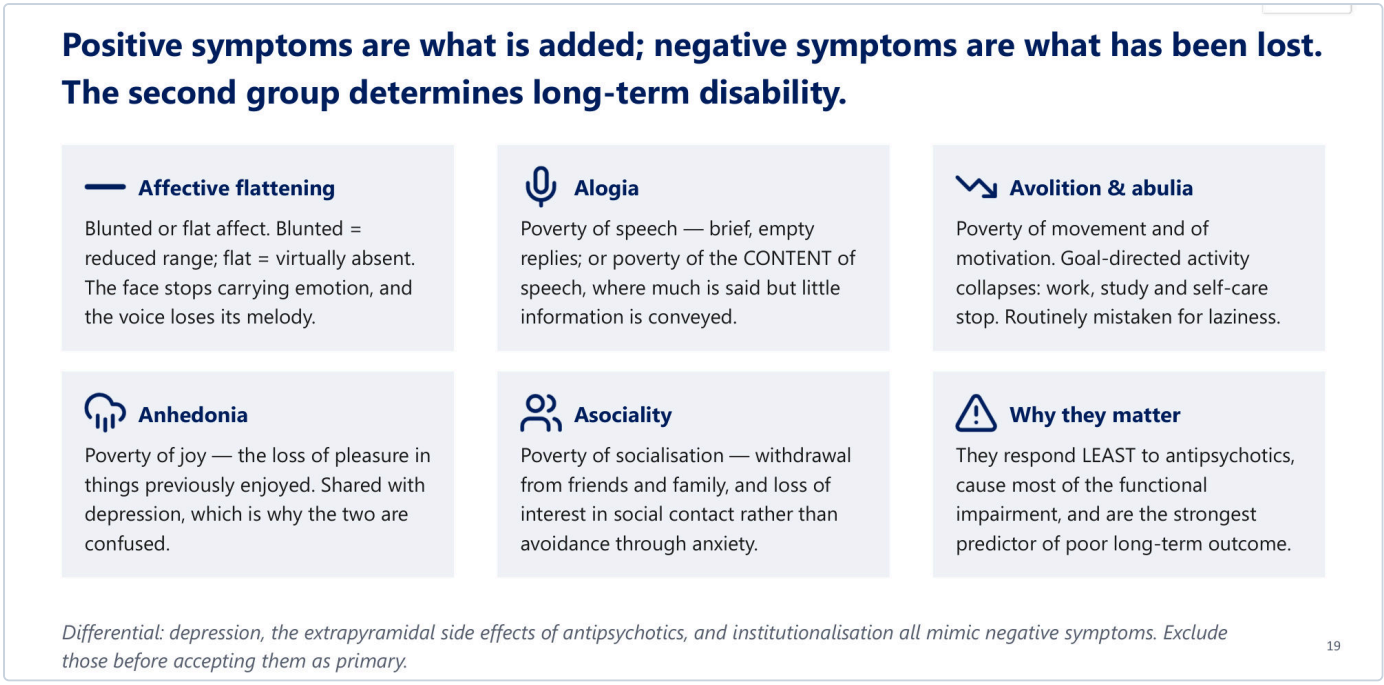


Slide 18: Disorganised behaviour and catatonia

2.9 Negative Symptoms: the Five A's (Slide 19) ⭐

A	Meaning
Affective flattening	Blunted (reduced range) or flat (virtually absent); face stops carrying emotion; voice loses its melody
Alogia	Poverty of speech (brief, empty replies) or poverty of **content** of speech
Avolition / abulia	Poverty of movement and motivation; work, study, self-care stop; **mistaken for laziness**
Anhedonia	Loss of pleasure; **shared with depression**
Asociality	Withdrawal from social contact through **loss of interest** (not anxious avoidance)

- ⭐ Negative symptoms **respond LEAST to antipsychotics**, cause **most of the functional impairment**, and are the **strongest predictor of poor long-term outcome**.
- **Differential (secondary negative symptoms): depression, antipsychotic EPS (parkinsonism), institutionalisation**. Exclude these first; they are **treatable**.



Slide 19: The five A's

2.10 Schneider's First-Rank Symptoms (Slide 20) ⭐

First-rank symptoms	Where they stand now
Thought **INSERTION**	Present in **~3 of 4** patients with schizophrenia
Thought **WITHDRAWAL**	**Also** in mania, psychotic depression, organic psychoses
Thought **BROADCASTING**	⭐ **DSM-5 REMOVED their special weight**: they no longer count double
3rd-person auditory hallucinations (voices discussing the patient)	Still a useful structured checklist
Running commentary; **made acts, feelings, impulses**	Their presence should **never end the search for an organic cause**

Mnemonic *[added]*: "ABCD" for first-rank symptoms:

- **A**uditory hallucinations (3rd person, commentary, *[added]* thought echo)
- **B**roadcasting / insertion / withdrawal of thought
- **C**ontrol (made acts, feelings, impulses; passivity)
- **D**elusional perception *[added; classic FRS not on the slide]*

⚠️ DSM-IV allowed a **single bizarre delusion or a running commentary** to satisfy Criterion A alone. ⭐ **DSM-5-TR abolished this: two symptoms are always required.**

> 🔁 **Recap: Part 1.**
>
> - Psychosis = a symptom (impaired reality testing).
> - Normal mimics: hypnagogic, bereavement, sleep deprivation.
> - 6 domains (positive 1–3 respond best; negative determine disability).
> - Delusion = fixed, false, **outside culture**. Non-bizarre → delusional disorder. Passivity = first-rank.
> - Hallucination is external with no stimulus; non-auditory → organic.
> - Catatonia is an emergency, commonest in mood disorders; lorazepam/ECT.
> - Negative symptoms = 5 A's.
> - FRS lost special weight in DSM-5.
>
> 🔗 **Link:** Fahad has persecutory + reference delusions, 3rd-person commentary, command voices and an 8-month negative prodrome.
>
> ▶️ **Up next: Exclude everything else before calling it schizophrenia.**

2.11 Delirium vs Primary Psychosis (Slides 22–24) ⭐⭐

Case (slides 22–23):

- 72-year-old, **day 3 after hemiarthroplasty**
- pulling at his cannula, sees **children in the room**, disoriented
- fully oriented this morning; briefly drowsy last night
- on **tramadol + oxybutynin**; **T 37.9 °C**
- "refer to psychiatry for acute psychosis"

→ ⭐ **B: Delirium, find the cause.** Five features make the diagnosis:

1. **ACUTE onset over hours**
2. **FLUCTUATION** across the day
3. **IMPAIRED consciousness and attention**
4. **DISORIENTATION**
5. **VISUAL hallucinations**

Also present:

- **two anticholinergic-burden drugs**
- **fever**

What to do:

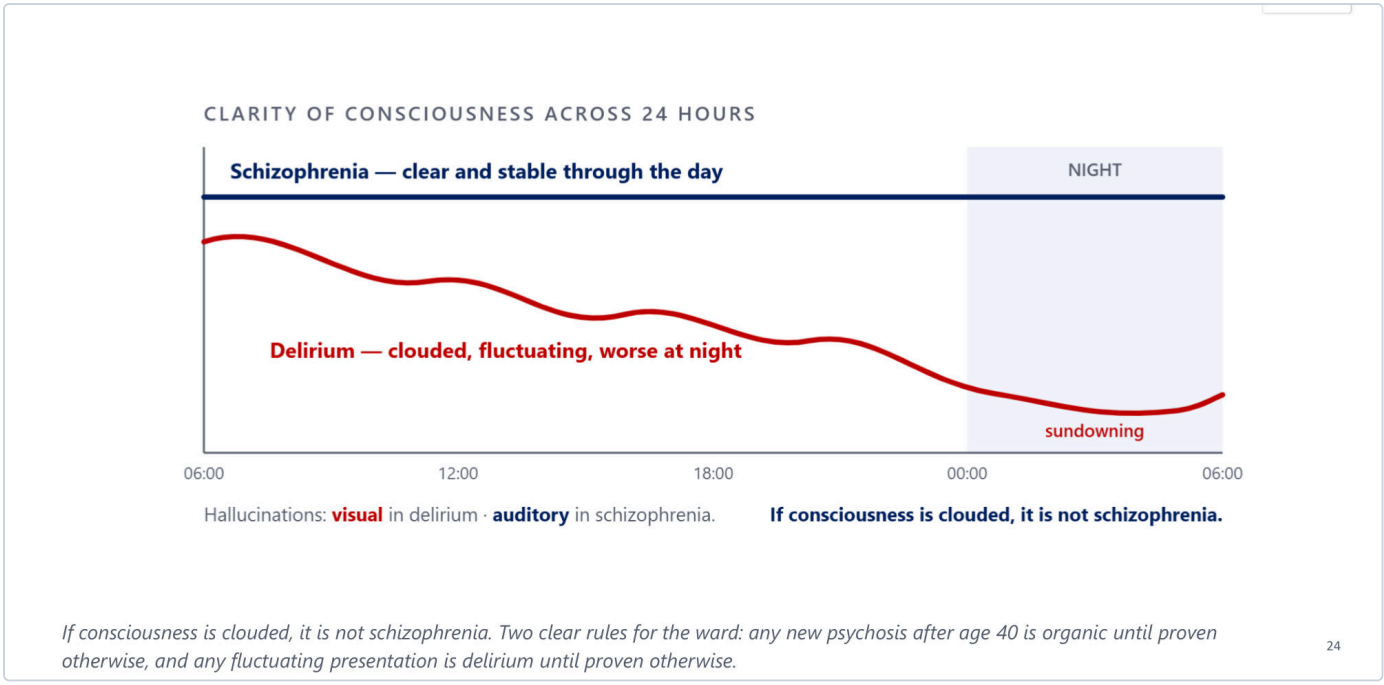
- examine
- screen for **infection and hypoxia**
- check **glucose, electrolytes, calcium, renal function**
- **review every drug**
- treat the cause

⭐ **Delirium in an elderly surgical patient carries a mortality comparable to MI.**

Slide 24 rules:

- ⭐ **If consciousness is clouded, it is not schizophrenia.**
- ⭐ **Any new psychosis after 40 = organic until proven otherwise.**
- ⭐ **Any fluctuating presentation = delirium until proven otherwise.**

- Hallucinations: **visual in delirium**, **auditory in schizophrenia**. Delirium is worse at night (**sundowning**).



Slide 24: Clarity of consciousness over 24 h, delirium vs schizophrenia

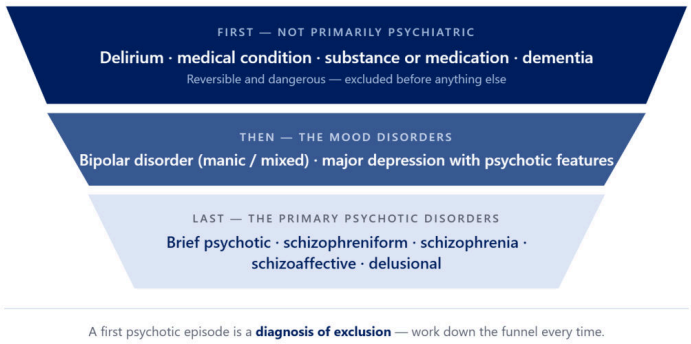
2.12 The Differential Funnel (Slide 25) ⭐

Work down the funnel **every time**:

1. **FIRST (not primarily psychiatric): delirium · medical condition · substance or medication · dementia**. Reversible and dangerous; excluded before anything else.
2. **THEN (mood disorders): bipolar (manic/mixed) · major depression with psychotic features**.
3. **LAST (primary psychotic disorders):** brief psychotic · schizophreniform · schizophrenia · schizoaffective · delusional.

⭐ **A first psychotic episode is a diagnosis of exclusion.**

Work down the funnel every time. The order is deliberate — the reversible and dangerous causes come first, and the label you were reaching for comes last.



A first psychotic episode is a diagnosis of exclusion. "Schizophrenia" is applied only after the tiers above it have been actively excluded, not assumed away.

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Slide 25: The differential funnel

2.13 Psychosis Due to a Medical Condition (Slide 26)

Group	Examples
CNS	Cerebrovascular disease · MS · tumour · Alzheimer, Parkinson · Huntington · **epilepsy (esp. temporal lobe)** · encephalitis · **neurosyphilis** · prion disease · neurosarcoidosis · **HIV/AIDS**
Endocrine	**Hyper/hypothyroidism · Cushing, Addison · hyper/hypocalcaemia · hypopituitarism**
Nutritional	**B12, folate, niacin (pellagra), thiamine** deficiency
Autoimmune	**SLE** · temporal arteritis · ⭐ **ANTI-NMDA RECEPTOR ENCEPHALITIS**
Metabolic / other	**Porphyria** · hepatic and uraemic encephalopathy · hypoglycaemia · hypoxia · **Wilson disease**

⭐ **Anti-NMDA receptor encephalitis:** in a **young patient**, psychosis plus **seizures, dyskinesias or autonomic instability**. It often presents first to psychiatry and is **treatable** (immunotherapy).

2.14 Substance- and Drug-Induced Psychosis (Slide 27) ⭐

Substances of misuse	⭐ Prescribed medication (the group students forget)
Amphetamine-type stimulants (incl. Captagon) and **cocaine** during intoxication	⭐ **CORTICOSTEROIDS**: dose-related; common on medical wards
Cannabis: dose-dependent; a risk factor for **later schizophrenia**	**Dopaminergic anti-parkinsonian** drugs (levodopa, dopamine agonists)
Hallucinogens (**LSD, MDMA**) and **PCP**	⭐ **ANTICHOLINERGICS and antihistamines**, esp. in the elderly
Alcohol, benzodiazepine, barbiturate WITHDRAWAL	Anticonvulsants; antimicrobials (**isoniazid, fluoroquinolones, mefloquine**)
Inhalants; toxins (**CO, organophosphates**)	**Methylphenidate, interferon, chemotherapy, digoxin**

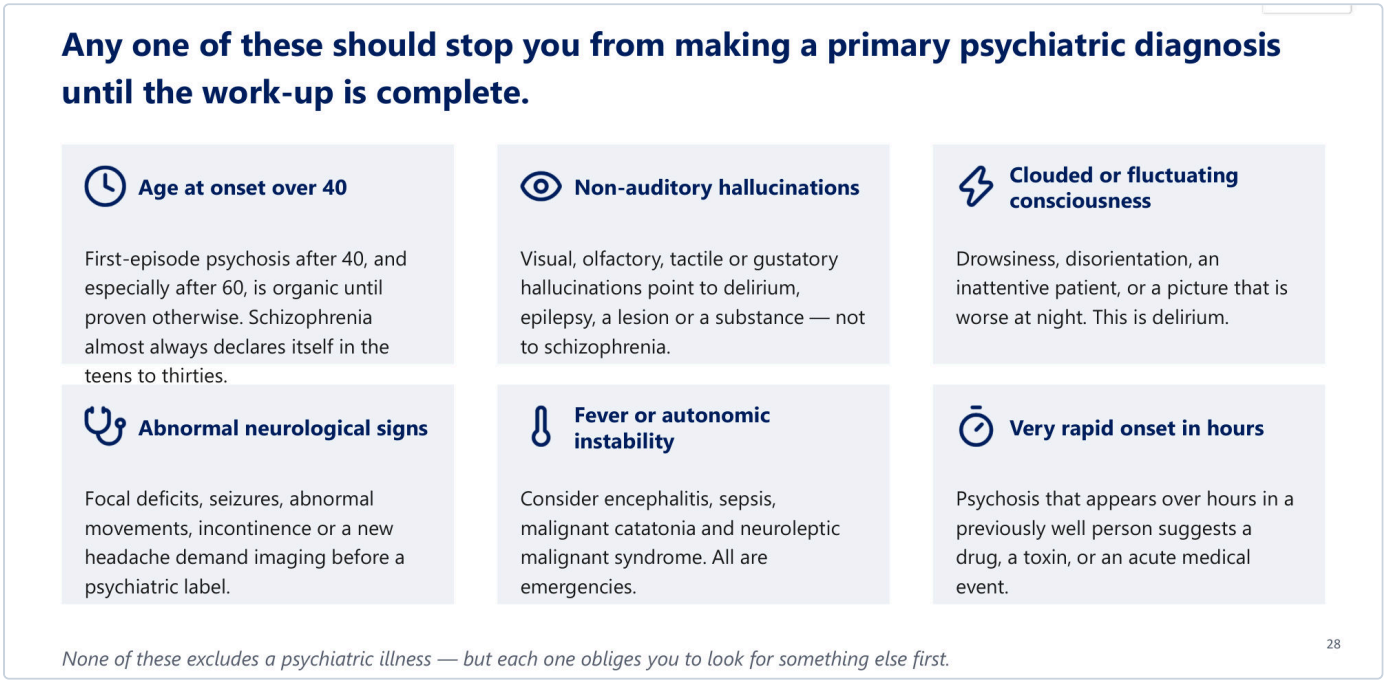
- Substance-induced psychosis typically **resolves within days–weeks of abstinence**.

- If it **persists** beyond intoxication/withdrawal, reconsider: stimulants can **unmask a primary psychotic illness**.

2.15 Red Flags: "This Is Not Schizophrenia" (Slide 28) ⭐⭐

Red flag	Why
Onset > 40 (esp. > 60)	Organic until proven otherwise; schizophrenia declares itself in the **teens–30s**
Non-auditory hallucinations	Delirium, epilepsy, lesion, substance
Clouded/fluctuating consciousness	Drowsy, disoriented, inattentive, worse at night = **delirium**
Abnormal neurological signs	Focal deficits, seizures, abnormal movements, incontinence, new headache → **imaging** first
Fever or autonomic instability	**Encephalitis, sepsis, malignant catatonia, NMS**: emergencies
Very rapid onset (hours)	Drug, toxin, acute medical event

None of these excludes psychiatric illness, but each one **obliges you to look for something else first**.



Slide 28: Red flags

2.16 First-Episode Work-Up (Slide 29) ⭐

Investigation	Looking for
⭐ **History, collateral, full physical examination**	**The single highest-yield step.** Never diagnose a first episode without collateral
CBC, renal, liver, electrolytes, calcium, glucose	Infection, uraemia, hepatic encephalopathy, Ca^{2+} disorders, hypoglycaemia
TSH, B12, folate	Common, cheap, reversible
⭐ **Urine drug screen**	Amphetamines, cannabis, cocaine, opioids: **required in every first episode**
HIV, syphilis serology; ESR/CRP, ANA if indicated	Infective/autoimmune (neurosyphilis, SLE)

Investigation	Looking for
CT/MRI brain; EEG if indicated	Tumour, stroke, demyelination; EEG for TLE or encephalitis
⭐ **Baseline before antipsychotics: weight, BMI, waist, BP, lipids, HbA1c, ECG**	You can't detect metabolic or QT harm later **without a baseline**

The **last row is the one most often skipped**: record it **before the first dose**.

Investigation	Looking for
History, collateral and full physical examination	The single highest-yield step — timeline, drugs, medical history, family account. Never diagnose a first episode without collateral
CBC, renal and liver profile, electrolytes, calcium, glucose	Infection, uraemia, hepatic encephalopathy, hyper/hypocalcaemia, hypoglycaemia
TSH, vitamin B12, folate	Thyroid disease and nutritional deficiency — common, cheap and reversible
Urine drug screen	Amphetamines, cannabis, cocaine and opioids — required in every first episode
HIV and syphilis serology; ESR/CRP, ANA if indicated	Infective and autoimmune causes, including neurosyphilis and SLE
CT or MRI brain; EEG if indicated	Tumour, stroke, demyelination; EEG for suspected temporal lobe epilepsy or encephalitis
Baseline before antipsychotics: weight, BMI, waist, BP, lipids, HbA1c, ECG	You cannot detect metabolic or QT harm later without a baseline recorded now

The last row is the one most often skipped, and the one you will regret. Record the baseline before the first dose.

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Slide 29: First-episode work-up

> 🔁 **Recap: Part 2.**
>
> - Clouded or fluctuating consciousness = delirium, not schizophrenia.
> - Funnel: organic → mood → primary psychotic.
> - Mimics: endocrine (thyroid, Cushing/Addison, Ca^{2+}), B12/thiamine, **anti-NMDA**, Wilson, porphyria.
> - Drugs: **steroids, anticholinergics**, levodopa, isoniazid, mefloquine; stimulants and cannabis.
> - Red flags: > 40, non-auditory, clouding, neuro signs, fever, hours.
> - Work-up includes **UDS** and a **metabolic + ECG baseline**.
>
> ▶️ **Up next: The psychotic disorders and the clock.**

2.17 The Psychotic Disorders: the Clock (Slides 31–34) ⭐⭐

Disorder	Duration	Symptoms	Function
⭐ **Brief psychotic disorder**	**1 day to < 1 month**	**≥ 1** of delusions, hallucinations, disorganised speech, disorganised/catatonic behaviour	**FULL return** to premorbid level
⭐ **Schizophreniform**	**1 to < 6 months**	**≥ 2** Criterion A	Dysfunction **NOT required**; full return if it resolves
⭐ **Schizophrenia**	**≥ 6 months** (≥ 1 month active)	**≥ 2** Criterion A	**No return** to premorbid level

Disorder	Duration	Symptoms	Function
⭐ **Delusional disorder** (off the clock)	**≥ 1 month** of delusions	Delusions **only**; no prominent hallucinations	**Preserved** outside the belief
⭐ **Schizoaffective** (off the clock)	**≥ 2 weeks** of psychosis **without** a mood episode	Mood episode concurrent with Criterion A; mood symptoms for **most of the illness**	—

Treatment does not wait for the label: all are treated as a **first psychotic episode from day one**.



Slide 31: The clock

Brief psychotic disorder (slide 32):

- often follows a **marked stressor**
- a **postpartum-onset specifier** exists
- recurrence and later reclassification are real, so **follow up**

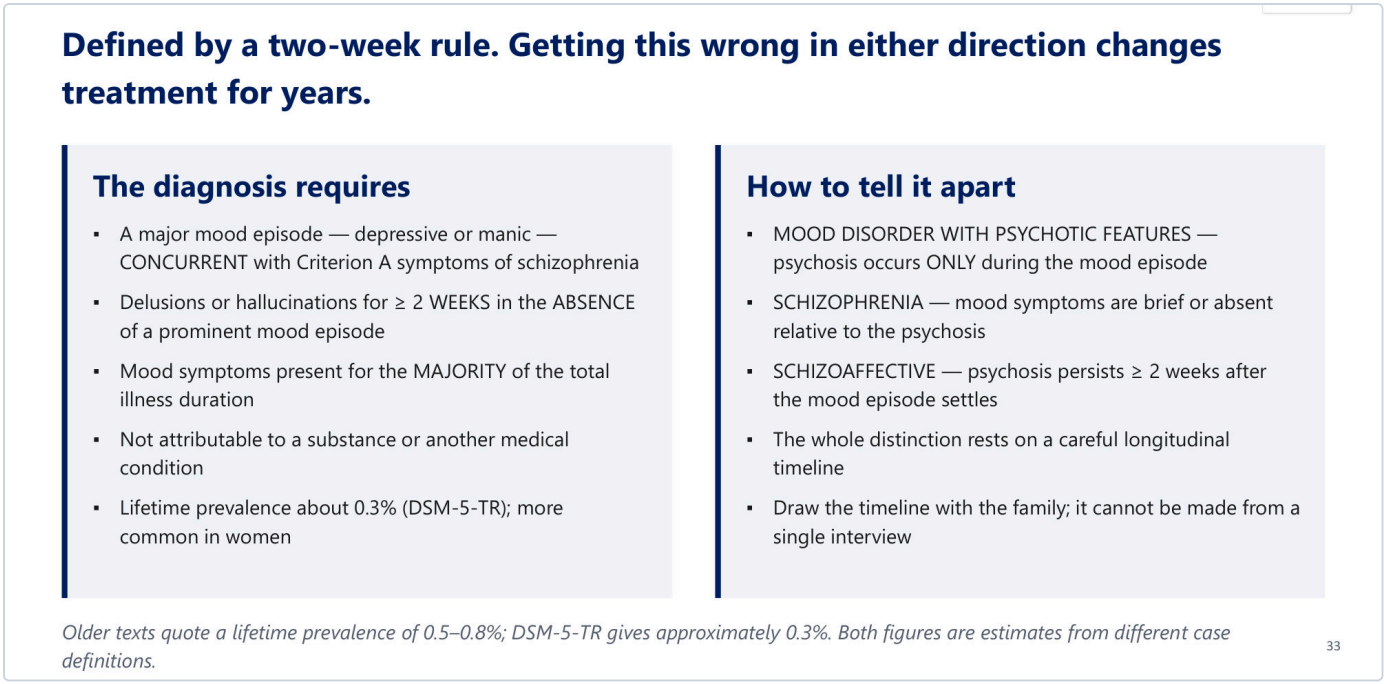
Schizophreniform (slide 32):

- identical to schizophrenia except **duration < 6 months** and **no dysfunction required**
- lifetime prevalence **~0.2%**; **M = F**
- if symptoms pass 6 months → **schizophrenia**

Schizoaffective Disorder: the Commonest Diagnostic Error (Slide 33) ⭐

Requires	How to tell it apart
A **major mood episode** (depressive or manic) **CONCURRENT** with Criterion A	**Mood disorder with psychotic features:** psychosis **ONLY during** the mood episode
⭐ **Delusions/hallucinations ≥ 2 WEEKS in the ABSENCE of a prominent mood episode**	**Schizophrenia:** mood symptoms **brief or absent** relative to psychosis
Mood symptoms for the **MAJORITY** of the illness	**Schizoaffective:** psychosis **persists ≥ 2 weeks after** the mood episode settles
Not due to substance/medical condition	The distinction rests on a **longitudinal timeline**, drawn **with the family**

Requires	How to tell it apart
Lifetime prevalence **~0.3%** (DSM-5-TR); **more common in women**	Older texts: 0.5–0.8% (different case definitions)



Slide 33: Schizoaffective disorder

Delusional Disorder (Slide 34)

Features	Subtypes and pitfalls
≥ 1 delusion for ≥ 1 month	**Persecutory · jealous · erotomanic · somatic · grandiose · mixed**
Typically **NON-BIZARRE**	Patients rarely present themselves; **family or police** bring them
No prominent hallucinations; Criterion A not met	**Insight absent** → engagement is the hardest part
⭐ **ENCAPSULATED**: rest of mental life intact	⭐ **Jealous subtype: documented risk of violence to the partner**
Behaviour not obviously odd; function preserved outside the delusion	⭐ **Somatic subtype** reaches **dermatology or surgery** long before psychiatry

These patients function normally, which is why the diagnosis is missed for years, and why **confrontation destroys the therapeutic relationship**.

> 🔁 **Recap: The clock.** < 1 month = brief (≥ 1 symptom, full recovery). 1–6 months = schizophreniform (≥ 2, dysfunction not required). ≥ 6 months = schizophrenia. Delusional disorder: ≥ 1 month, non-bizarre, encapsulated. Schizoaffective: psychosis ≥ 2 weeks without mood + mood for most of the illness.
>
> 🔗 **Link:** Q2 (29-year-old woman, voices continuing 6 weeks after depression settled) = schizoaffective.
>
> ▶️ **Up next: Schizophrenia itself.**

2.18 Epidemiology of Schizophrenia (Slide 39) ⭐

Measure	Figure	Comment
Lifetime prevalence	⭐ **≈ 1%** (classical)	DSM-5-TR: **~0.3–0.7%**
Onset: men	⭐ **≈ 21 years**	Late teens–mid-20s; onset before puberty or after 45 is rare
Onset: women	⭐ **≈ 27 years**	**Second, smaller peak around menopause**
Sex ratio	**M ≥ F**	Slight male excess
Male course	**More severe**	Earlier onset, more **negative** symptoms, poorer response and prognosis
Female course	**Less severe**	Later onset, more **positive** symptoms, better response and prognosis
Life expectancy	⭐ **15–20 years shorter**	Mainly **cardiovascular/metabolic disease and smoking**, not suicide alone

2.19 Stress–Diathesis Model (Slide 40)

Diathesis: the vulnerability	Stress: the trigger
Genetic loading (the largest single contribution)	**Substance use**, esp. **cannabis and stimulants in adolescence**
Dopaminergic, serotonergic, glutamatergic dysfunction	**Bereavement**, loss of a significant relationship
Neurodevelopmental insult: obstetric complications, maternal infection, **winter birth**	**Migration, urban upbringing**, social adversity
Structural/functional brain changes **before onset**	Major role change: leaving home, **starting university**
Present long before any symptom	⭐ **High expressed emotion at home**: a potent trigger for **RELAPSE**

Neither column alone is sufficient. Explain it to families this way: **the vulnerability was there; the stressor was the last push, not the cause.**

2.20 Genetics (Slide 41) ⭐

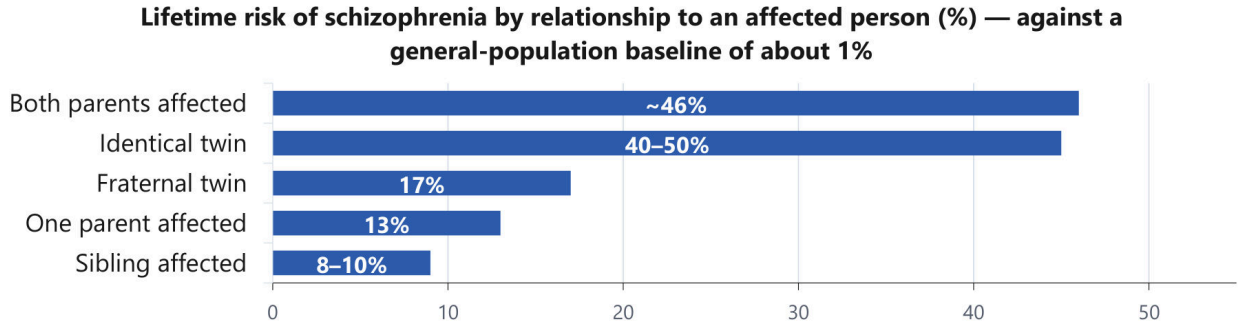
- ⭐ **50–70%** of the risk is genetic.
- **Identical twin concordance 40–50%**, so **environment must matter**.
- **General population ≈ 1%.**

Relationship to an affected person	Lifetime risk
Both parents	⭐ **~46%**
Identical twin	⭐ **40–50%**
Fraternal twin	**17%**
One parent	⭐ **13%**
Sibling	**8–10%**

50–70%
of the risk of developing schizophrenia is attributed to genetic factors

40–50%
concordance in identical twins — so genes are not the whole story

~1%
baseline lifetime risk in the general population



Where a source gives a range, the bar is drawn to the middle of it and the LABEL carries the full published range. The decisive observation is the identical-twin figure: genetically identical people are concordant less than half the time, so environment must matter. Source: Kaplan & Sadock's Synopsis of Psychiatry.

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Slide 41: Genetic risk by relationship

2.21 The Dopamine Hypothesis: Four Pathways (Slides 42, 45–46) ⭐⭐

Pathway	State in schizophrenia	Effect of D2 blockade
⭐ **Mesolimbic**	**Hyperdopaminergic → POSITIVE symptoms**	**The therapeutic effect**
Mesocortical	**Under-active → negative and cognitive symptoms**	Blocking further can **WORSEN** these
Nigrostriatal (motor)	—	⭐ **EPS and tardive dyskinesia**
Tuberoinfundibular (endocrine)	—	⭐ **↑ Prolactin**: galactorrhoea, amenorrhoea, sexual dysfunction (± ↓ bone density)

Pharmacological proof:

- **cocaine and amphetamine** raise dopamine and **exacerbate** psychosis
- antipsychotics **block D2** and reduce it

Mnemonic *[added]*: meso**LIMBIC** = **L**oud (positive) · meso**CORTICAL** = **C**old (negative/cognitive) · nigro**STRIATAL** = **S**tiff (EPS) · **TUBERO**infundibular = **T**eat (prolactin).

Case (slides 45–46): 24-year-old woman, 3 weeks on **haloperidol**: voices almost gone, now **amenorrhoea + bilateral galactorrhoea**, pregnancy test negative, prolactin markedly raised.

- → ⭐ **D: tuberoinfundibular.** Hypothalamic dopamine **tonically INHIBITS prolactin**.
- The improvement in voices = the **mesolimbic** effect of the same drug.
- **Manage:**
  - reduce the dose, or switch to a **prolactin-sparing agent (aripiprazole)**
  - don't just stop treatment, but don't ignore it: it is a common, **unspoken reason patients stop medication**

Excess dopamine in the mesolimbic pathway produces the positive symptoms. Blocking D2 receptors treats them — and produces the side effects, in the other three pathways.

Mesolimbic — the target

- Hyperdopaminergic state → POSITIVE symptoms
- D2 blockade here is the therapeutic effect

Mesocortical — under-active

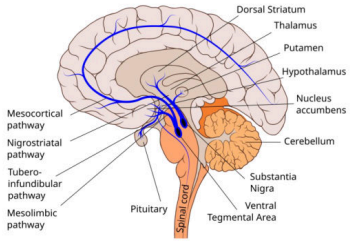
- Low dopamine → negative and cognitive symptoms
- Blocking it further can WORSEN these

Nigrostriatal — motor

- D2 blockade → extrapyramidal side effects and tardive dyskinesia

Tuberoinfundibular — endocrine

- D2 blockade → raised prolactin: galactorrhoea, amenorrhoea, sexual dysfunction



Dopaminergic pathways. Wikimedia Commons — User:Slashme, after Patrick J. Lynch. CC BY-SA 4.0.

The pharmacological proof: cocaine and amphetamine raise dopamine and EXACERBATE psychosis; first- and second-generation antipsychotics block D2 and REDUCE it.

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Slide 42: The four dopamine pathways

2.22 Beyond Dopamine (Slides 43–44)

Serotonin & glutamate	Neurodevelopmental	Brain changes	Autoimmune
5-HT2A antagonism underlies SGAs	**Obstetric complications, maternal influenza, famine**	⭐ **Enlarged lateral ventricles**; ↓ total brain volume	Raised inflammatory markers in a **subgroup**
⭐ **NMDA receptor HYPOFUNCTION**: the leading alternative model	Excess **winter/early-spring births**	Accelerated grey-matter loss in adolescence	**Anti-NMDA encephalitis** = proof of principle (antibody psychosis responding to immunotherapy)
Ketamine and PCP block NMDA → reproduce **positive, negative AND cognitive** symptoms	Premorbid cognitive/motor abnormalities in childhood	↓ hippocampal, amygdala, temporal volume	Suspect with seizures, dyskinesias, autonomic instability
Best pharmacological model of **negative** symptoms	Lesion laid down early; symptoms emerge in adolescence (**explains the timing**)	⭐ **HYPOFRONTALITY**; present **before** antipsychotic exposure	Not a routine screen, but not to miss

⭐ **None of these is diagnostic. Imaging in schizophrenia is done to EXCLUDE other causes, not to confirm.**

🔁 **Recap: Aetiology.**

- Stress–diathesis; genes 50–70% (MZ 40–50%, one parent 13%, both 46%, sibling 8–10%).
- Dopamine: mesolimbic (positive, the target), mesocortical (negative), nigrostriatal (EPS), tuberoinfundibular (prolactin).
- NMDA hypofunction (ketamine/PCP); neurodevelopmental (winter birth, obstetric); ventricles enlarged, hypofrontality.
- High expressed emotion → relapse.

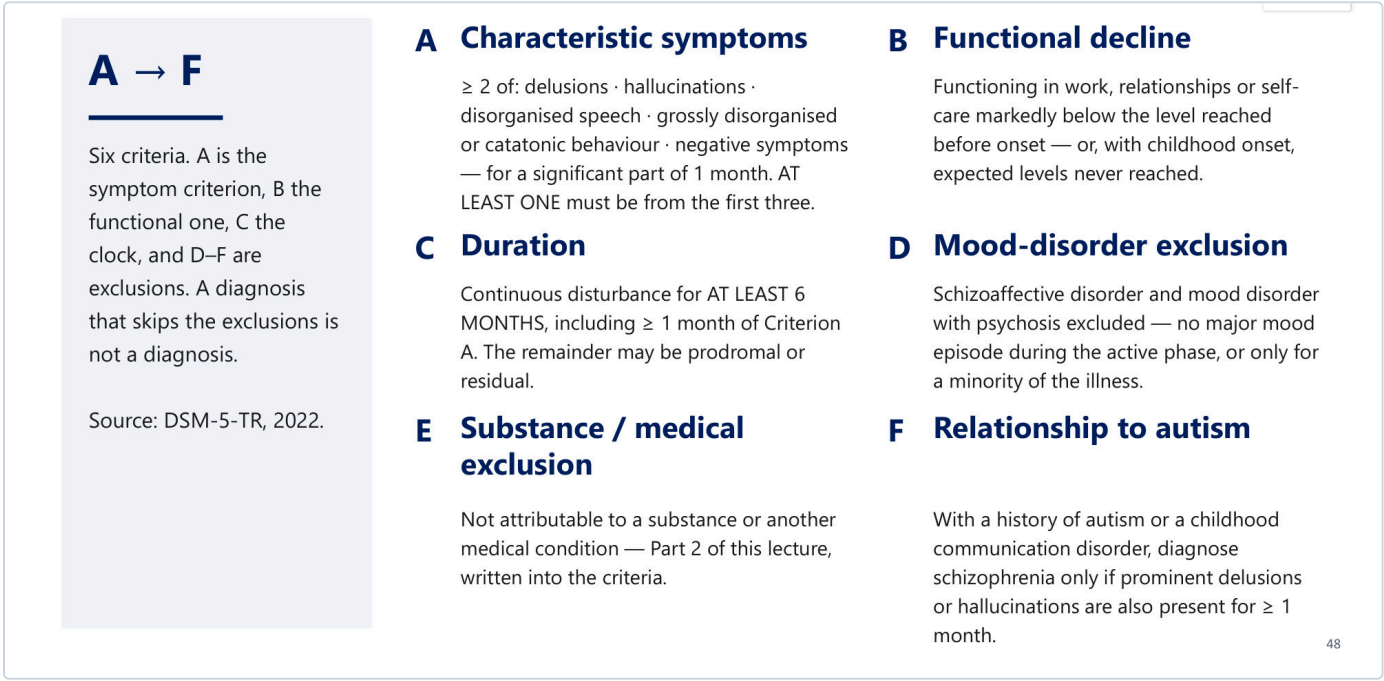
▶️ **Up next: DSM-5-TR criteria.**

2.23 DSM-5-TR Criteria for Schizophrenia (Slides 48–49) ⭐⭐

	Criterion	Content
A	**Characteristic symptoms**	**≥ 2** of: **delusions · hallucinations · disorganised speech** · grossly disorganised/catatonic behaviour · negative symptoms, for a significant part of **1 month**. ⭐ **AT LEAST ONE from the first three**
B	**Functional decline**	Work, relationships or self-care **markedly below** the pre-onset level (childhood onset: expected level not reached)
C	**Duration**	⭐ **Continuous ≥ 6 MONTHS**, including **≥ 1 month of Criterion A**; the rest may be prodromal/residual
D	**Mood-disorder exclusion**	Schizoaffective and mood disorder with psychosis excluded
E	**Substance/medical exclusion**	Part 2 of this lecture, written into the criteria
F	**Relationship to autism**	With autism/childhood communication disorder history, diagnose only if prominent **delusions or hallucinations ≥ 1 month**

The rule inside the rule (slide 49):

- ⭐ **Negative symptoms alone can never make the diagnosis.**
- **Catatonia (4) + negative symptoms (5) together do NOT satisfy Criterion A.**
- **Delusions + negative symptoms DO** satisfy it.
- < 1 month of symptoms **treated successfully** still counts.
- The DSM-IV **single-bizarre-delusion shortcut was ABOLISHED**.



Slide 48: DSM-5-TR criteria A–F

2.24 Positive, Negative and Cognitive Symptoms (Slides 50–52)

Group	Content	Response
Positive (added)	Delusions (persecutory commonest); **auditory** hallucinations (3rd person, commentary; ask about **command**); disorganised speech; disorganised behaviour; catatonia	⭐ **Best**; improve within **2–4 weeks**

Group	Content	Response
Negative (lost)	**Five A's**; often **precede the first positive symptom by years**	Poor. Exclude **secondary** causes (depression, drug-induced parkinsonism, understimulation)
Cognitive	**Attention**; **working memory/multitasking**; ⭐ **executive function: "SOAP"** (**S**equencing, **O**rganisation, **A**bstracting, **P**lanning); concrete proverbs	Present **premorbidly**, persist between episodes. **No drug reliably helps**; **cognitive remediation and supported employment** have evidence

⭐ **Cognitive impairment predicts functional outcome (employment, independent living) better than positive symptoms do.** This is why a patient whose voices have resolved may still be unable to return to university.

2.25 Historical Subtypes (Slide 53): Removed in DSM-5, Still Examined

Subtype	Defining feature
Catatonic	≥ 1 catatonic symptom, regardless of anything else
Disorganised (hebephrenic)	Disorganised speech and behaviour, flat/inappropriate affect, **no catatonia**
Paranoid	**Delusions and hallucinations dominate**; no catatonia or disorganisation; ⭐ **best preserved function**
Undifferentiated	Criterion A met; fits none of the above
Residual	No prominent positive symptoms; attenuated negative symptoms and odd beliefs persist

Why abolished: they were **unstable over time**, overlapped heavily, and **didn't predict treatment response or outcome**. They were replaced by **dimensional severity ratings**.

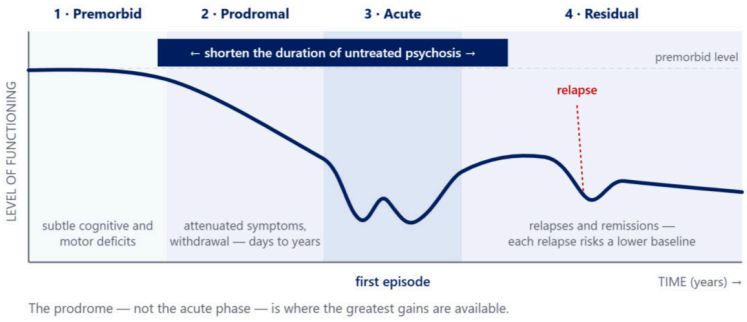
> 🔁 **Recap: Diagnosis.**
> - A: ≥ 2 for 1 month, ≥ 1 from delusions/hallucinations/disorganised speech.
> - B: decline. C: ≥ 6 months. D–F: exclusions (mood, substance/medical, autism).
> - Positive respond; negative and cognitive determine function.
> - Subtypes are gone (paranoid had the best function).
>
> ▶️ **Up next: Course and prognosis.**

2.26 Course: Four Phases (Slides 55–56)

Phase	Features
1. Premorbid	Subtle **cognitive and motor deficits**
2. Prodromal	**Attenuated symptoms, withdrawal**; days to years. ⭐ **Shortening the duration of untreated psychosis (DUP)** is one of the few things that **measurably improves outcome**
3. Acute (active)	≥ 2 Criterion A symptoms; often follows a stressor or substance use; severe impairment; usually **hospitalisation**; when patients first present
4. Residual	Attenuated positive/negative symptoms; **depression and cognitive impairment common**; function below premorbid; **relapse is the main risk**

⭐ A chronic illness of relapses and remissions: **each relapse risks a lower functional baseline.**

Recognising the prodrome matters: shortening the duration of untreated psychosis is one of the few things that measurably improves outcome.



Long duration of untreated psychosis predicts worse outcome — which is why the prodromal phase, not the acute one, is where the greatest gains are available.

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Slide 55: Phases and the duration of untreated psychosis

2.27 Physical Health and the Mortality Gap (Slide 57) ⭐

Domain	Detail
Smoking	~**3× more likely** to smoke, and more heavily; ⭐ **largest single contributor to premature death**
Metabolic	Obesity, **T2DM**, dyslipidaemia, CVD (illness + lifestyle + medication)
Infection	HIV and blood-borne infection (risk behaviour, poor preventive care)
Substance use	Alcohol and drugs: worse adherence, relapse, physical health
Suicide	⭐ **~5%** (Palmer 2005: ≈ 4.9%; older texts **10–13%**). Highest **early in illness, young men, those with insight and depression**
⭐ **Diagnostic overshadowing**	New physical symptoms **attributed to the psychiatric illness** and not investigated

2.28 Prognosis (Slides 58–59) ⭐

Outcome split:

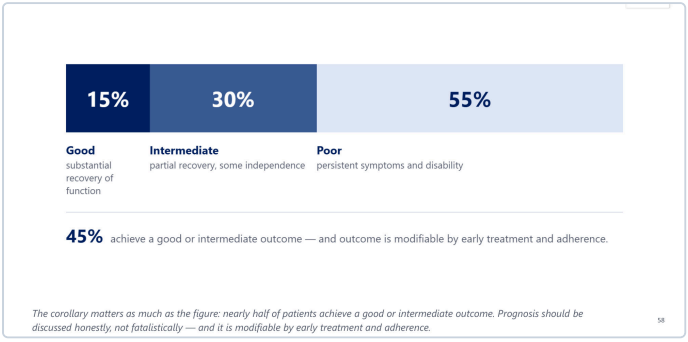
- ⭐ **15% good** (substantial recovery)
- **30% intermediate** (partial recovery, some independence)
- **55% poor** (persistent symptoms and disability)

→ **45% achieve good/intermediate**; outcome is **modifiable** by early treatment and adherence.

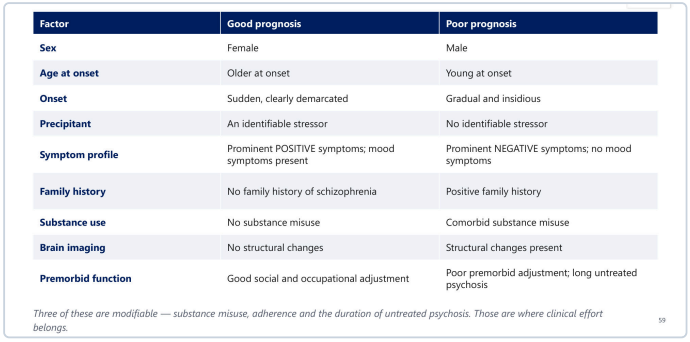
Factor	Good prognosis	Poor prognosis
Sex	**Female**	Male
Age at onset	**Older**	Young
Onset	**Sudden**, clearly demarcated	Gradual, insidious
Precipitant	**Identifiable stressor**	None
Symptoms	**Positive**; **mood symptoms present**	**Negative**; no mood symptoms
Family history	None	Positive

Factor	Good prognosis	Poor prognosis
Substance use	None	Comorbid misuse
Brain imaging	No structural change	Structural changes
Premorbid function	**Good** social/occupational adjustment	Poor adjustment; **long DUP**

⭐ **Three modifiable factors:** substance misuse, adherence, duration of untreated psychosis.



Slide 58: Prognosis 15/30/55



Slide 59: Prognostic factors

> 🔁 **Recap: Course.**
>
> - Premorbid → prodrome (shorten the DUP) → acute → residual.
> - Life expectancy −15–20 years (smoking is the biggest driver); suicide ~5% (older texts 10–13%).
> - 15/30/55.
> - Good prognosis: female, late and sudden onset, a stressor, positive and mood symptoms, good premorbid function.
>
> ▶️ **Up next: Treatment.**

2.29 Management Framework (Slide 61)

- **Four strands, delivered together:**
  1. **pharmacological** (antipsychotics)
  2. **psychological** (CBT for psychosis, supportive therapy)
  3. **family** (psychoeducation, expressed emotion)
  4. **social** (housing, income, occupational rehabilitation)
- **Psychoeducation** for patient and family covers:
  - a **brain illness, not a weakness**
  - medication effects, onset, side-effects
  - ⭐ **written, patient-specific EARLY SIGNS OF RELAPSE**
  - ⭐ **HIGH EXPRESSED EMOTION** (criticism, hostility, over-involvement) **drives relapse**, and family intervention can change it
  - where to get urgent help

2.30 Acute Phase (Slide 62)

Assess and protect	Treat and engage
Diagnosis: history, MSE, examination, investigations	Plan treatment **with** the patient
⭐ **PREVENT HARM**: risk to self/others; admit if needed	Reduce symptom severity as quickly as is safe

Assess and protect	Treat and engage
Control disturbed behaviour, **de-escalation before medication**	Aim for rapid return toward optimal functioning
⭐ **COLLATERAL**: essential, not optional	Alliance with patient **and** family
Address precipitants: substances, non-adherence	**Psychoeducation early**, not at discharge

Involuntary admission: governed by the **Saudi Mental Health Law**; a last resort; **document the reasoning**.

2.31 First- vs Second-Generation Antipsychotics (Slide 63) ⭐⭐

	First generation (typical)	Second generation (atypical)
Mechanism	**D2 blockade**	**D2 + 5-HT2A antagonism**
Examples	**Haloperidol, chlorpromazine, fluphenazine, trifluoperazine**	**Risperidone, olanzapine, quetiapine, aripiprazole, paliperidone, clozapine**
Positive symptoms	Effective	Effective; ⭐ **no clear superiority except clozapine**
Negative & cognitive	Little effect; may worsen	Modest benefit at best
EPS	⭐ **HIGH** (dystonia, akathisia, parkinsonism, TD)	Lower, but dose-dependent, **not absent**
Metabolic	Lower (**chlorpromazine** is an exception)	⭐ **HIGH**: weight, diabetes, lipids; **worst with olanzapine and clozapine**
Prolactin	Raised	Raised with **risperidone, paliperidone**; ⭐ **aripiprazole spares it**

⭐ Apart from clozapine, the generations are **similarly effective** (CATIE, CUtLASS). **Choose on side-effect profile, patient preference, previous response and cost.**

	First generation (typical)	Second generation (atypical)
Mechanism	Predominantly D2 blockade	D2 blockade plus 5-HT2A antagonism
Examples	haloperidol, chlorpromazine, fluphenazine, trifluoperazine	risperidone, olanzapine, quetiapine, aripiprazole, paliperidone, clozapine
Positive symptoms	Effective	Effective — no clear superiority except clozapine
Negative & cognitive	Little effect; may worsen	Modest benefit at best
Extrapyramidal effects	HIGH — dystonia, akathisia, parkinsonism, tardive dyskinesia	Lower, but dose-dependent and not absent
Metabolic effects	Lower (chlorpromazine is an exception)	HIGH — weight gain, diabetes, dyslipidaemia; worst with olanzapine and clozapine
Prolactin	Raised	Raised with risperidone and paliperidone; aripiprazole spares it

The honest summary: apart from clozapine, the two generations are similarly effective. The choice is made on SIDE-EFFECT profile, patient preference, previous response and cost — not on presumed superiority.

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Slide 63: FGA vs SGA

2.32 Prescribing Principles (Slide 64)

Rules	Timing and choice
Medication is **essential but not the whole** of treatment	Agitation settles in **days**; psychosis over ⭐ **2–6 weeks**
Individualise	⭐ **Adequate trial = 4–6 weeks at a therapeutic dose** before switching
⭐ **Start low, go slow**; lowest effective dose	Choose on **previous response, a family member's response**, side-effects
⭐ **AVOID POLYPHARMACY**: one antipsychotic at a time	Interactions, cost, availability, **patient preference**
Simple regimens succeed	**Involve the patient**: the strongest adherence intervention

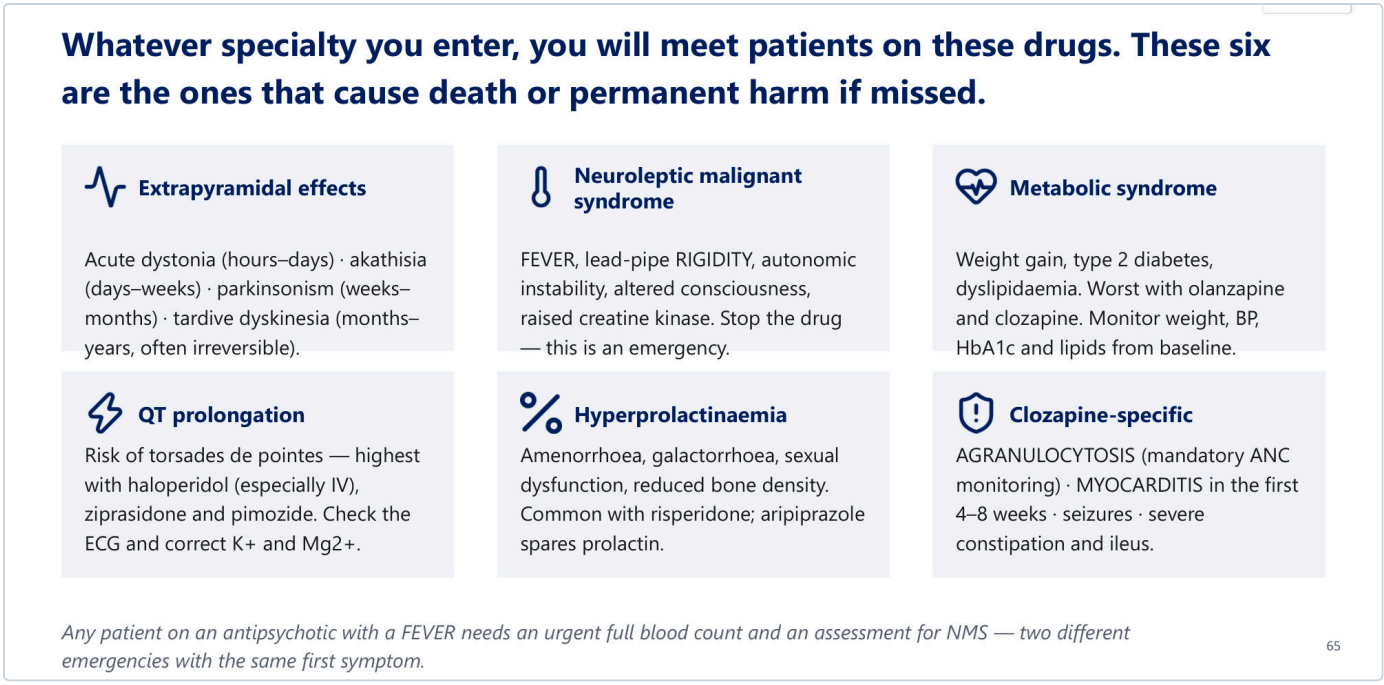
Commonest errors:

- switching too early
- **adding a second antipsychotic** instead of optimising the first
- never asking what the patient can tolerate

2.33 Harms You Must Recognise (Slides 65–67) ⭐⭐

Harm	Key points
EPS	Dystonia (hours–days) · akathisia (days–weeks) · parkinsonism (weeks–months) · TD (months–years, often irreversible)
⭐ **NMS**	**FEVER, lead-pipe RIGIDITY, autonomic instability, altered consciousness, ↑ CK**. **Stop the drug**; emergency
Metabolic syndrome	Weight, T2DM, lipids; worst **olanzapine/clozapine**. Monitor weight, BP, HbA1c, lipids **from baseline**
⭐ **QT prolongation**	Torsades; highest with **haloperidol (esp. IV), ziprasidone, pimozide**. ECG; correct **K^+ and Mg^{2+}**
Hyperprolactinaemia	Amenorrhoea, galactorrhoea, sexual dysfunction, ↓ bone density. Common with **risperidone**; **aripiprazole spares**
⭐ **Clozapine-specific**	**Agranulocytosis** (mandatory ANC monitoring) · **myocarditis in weeks 4–8** · seizures · **severe constipation/ileus**

⭐ **Any patient on an antipsychotic with FEVER needs an urgent FBC and an NMS assessment.** These are two different emergencies with the same first symptom.



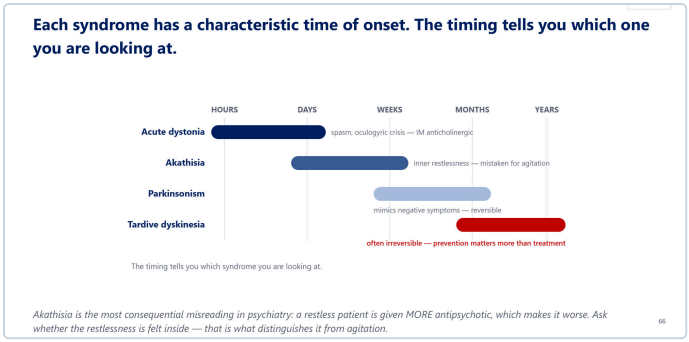
Slide 65: Six harms

EPS by the Clock (Slides 66–67) ⭐

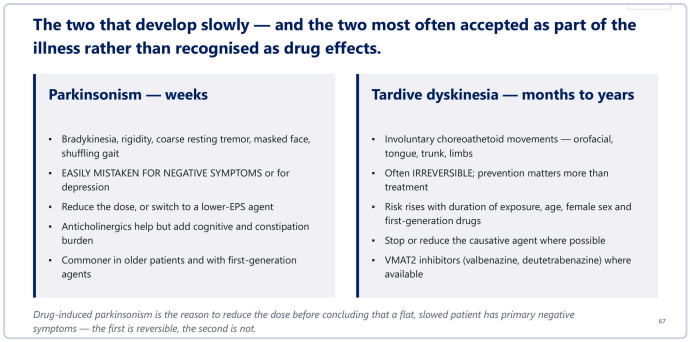
Syndrome	Onset	Features	Management
⭐ **Acute dystonia**	**Hours–days**	Muscle spasm, **oculogyric crisis**, torticollis	⭐ **IM anticholinergic** (*[added]* procyclidine/benztropine)
⭐ **Akathisia**	**Days–weeks**	**Inner restlessness**, pacing, can't sit	**Reduce dose or add propranolol**. ⭐ **Do NOT increase** the antipsychotic
Parkinsonism	**Weeks–months**	Bradykinesia, rigidity, coarse resting tremor, masked face, shuffling gait. ⭐ **Mimics negative symptoms/depression**; **reversible**	Reduce dose / lower-EPS agent; anticholinergics (cognitive and constipation burden). Commoner in the **elderly** and with **FGAs**
⭐ **Tardive dyskinesia**	**Months–years**	Involuntary **choreoathetoid orofacial/tongue**, trunk, limb movements. ⭐ **Often IRREVERSIBLE**	Prevention > treatment. Stop/reduce the agent; **VMAT2 inhibitors (valbenazine, deutetrabenazine)**. Risk ↑ with duration, **age, female sex, FGAs**

- ⭐ **Akathisia is the most consequential misreading in psychiatry:** a restless patient is given more antipsychotic, which makes it worse. Ask whether the restlessness is **felt inside**.
- Reduce the dose **before** concluding a flat, slowed patient has primary negative symptoms: parkinsonism is reversible; negative symptoms are not.

Mnemonic *[added]* for EPS timing: "4 hours, 4 days, 4 weeks, 4 months": dystonia, akathisia, parkinsonism, tardive dyskinesia (approximate).



Slide 66: EPS by the clock



Slide 67: Parkinsonism and TD

2.34 Clozapine and Treatment Resistance (Slides 68–70) ⭐⭐

When indicated	What it demands
⭐ **Treatment-resistant schizophrenia: failure of ≥ 2 adequate trials**	⭐ **Mandatory neutrophil monitoring** (agranulocytosis is the reason for the register)
Adequate trial = therapeutic dose for 4–6 weeks with confirmed adherence	⭐ **MYOCARDITIS in weeks 4–8**: fever, chest pain, dyspnoea, ↑ troponin
⭐ **Persistent suicidality** in schizophrenia	**Seizures**: dose-related; **myoclonus** may precede them
Severe tardive dyskinesia	⭐ **Constipation → ILEUS**: a recognised cause of death
Underused; often started years late	Sedation, **hypersalivation**, marked weight gain

- The **only antipsychotic with proven superior efficacy**.
- ⭐ **Never restart clozapine after > 48 h interruption without re-titrating from a low dose.** Sudden resumption can cause **collapse**; a common error on non-psychiatric wards.

Case (slides 69–70):

- 31-year-old, **5 weeks on clozapine**
- **2 days of fever 38.6 °C**, malaise, mild chest discomfort, **HR 118**, chest clear
- triage: "viral illness, reassure and discharge"

→ ⭐ **B: FBC, ECG, troponin, CRP; call psychiatry.** Three clozapine emergencies present with fever:

1. **agranulocytosis**: urgent FBC/neutrophils
2. **myocarditis**: weeks 2–8, fever, tachycardia, chest discomfort, ↑ troponin, CRP, **eosinophils** (this man has both)
3. **NMS**: rigidity, autonomic instability, altered consciousness, ↑ CK

Discharging him as "viral" = **diagnostic overshadowing**.

The only antipsychotic with proven superior efficacy — and the one with the most demanding safety requirements.

When it is indicated

- TREATMENT-RESISTANT schizophrenia: failure of ≥ 2 adequate antipsychotic trials
- An adequate trial = a therapeutic dose for 4–6 weeks with confirmed adherence
- Also indicated for persistent suicidality in schizophrenia
- And for severe tardive dyskinesia
- Underused and often started years later than it should be

What it demands

- MANDATORY neutrophil monitoring — agranulocytosis is the reason for the register
- MYOCARDITIS in the first 4–8 weeks: fever, chest pain, dyspnoea, raised troponin
- Seizures — dose-related; myoclonus can precede them
- Severe constipation progressing to ILEUS, a recognised cause of death
- Sedation, hypersalivation and marked weight gain

Never restart clozapine after an interruption of more than 48 hours without re-titrating from a low dose — sudden resumption at the previous dose can cause collapse. This is a common error when a patient is admitted to a non-psychiatric ward.

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Slide 68: Clozapine

2.35 Stabilisation, Maintenance and Psychosocial Treatment (Slides 71–72)

Stabilisation	Maintenance
Consolidate response, prevent early relapse	Improve function and quality of life
Adapt to community life; **reduce stress**	Sustain remission; **PREVENT RELAPSE**
⭐ **Monitor side-effects actively**: this is when patients quietly stop treatment	Monitor physical health: **weight, HbA1c, lipids, ECG**
Begin psychosocial intervention	Reduce modifiable stress (finances, housing, work); address substances and family conflict

Psychosocial intervention	Target
CBT for psychosis	The **distress and conviction** attached to residual voices/delusions (not arguing them away); for **persistent** symptoms
Supportive therapy	Long-term engagement
⭐ **Family intervention**	Psychoeducation + **expressed emotion**; among the **best-evidenced relapse prevention**
Group therapy / social skills	Asociality and withdrawal
⭐ **Occupational rehabilitation (supported employment)**	Work is protective; inactivity deepens negative symptoms
Social and community support	Housing, income, case management

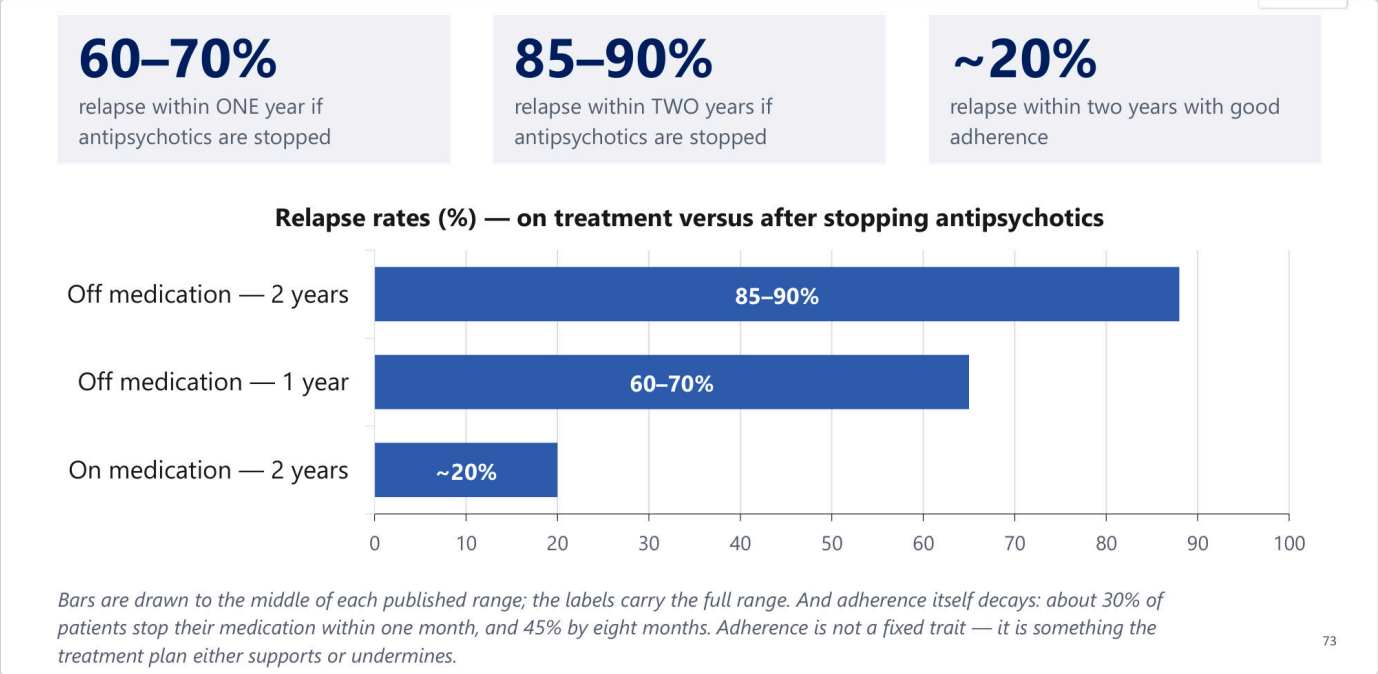
⭐ **Family intervention and supported employment** have the **strongest functional evidence**, and both are under-provided.

2.36 Relapse and Adherence (Slides 73–74) ⭐

Situation	Relapse rate
⭐ **Off medication, 1 year**	**60–70%**
⭐ **Off medication, 2 years**	**85–90%**
On medication with good adherence, 2 years	**~20%**

- **Adherence decays: ~30% stop within 1 month, 45% by 8 months.**
- **The three levers:**
  1. ⭐ **ADHERENCE**: the single most important factor
  2. strong **psychosocial support**
  3. **ABSTINENCE** (cannabis, stimulants)

Plus a **written relapse plan** with the patient's own early warning signs, and **continuity** of team. - **Why patients stop:** - **side-effects** (weight, sedation, sexual dysfunction) - lack of insight / feeling well - lack of efficacy / complex regimen - **cost, distance** - poor rapport; substance misuse - ⭐ **Consider a LONG-ACTING INJECTABLE (LAI) early**, especially after a relapse due to non-adherence, not as a last resort.



Slide 73: Relapse on vs off medication

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3. High-Yield Comparison Tables

Table A: The Psychotic Disorders

Disorder	Duration	Min. symptoms	Key feature
Brief psychotic	1 day – < 1 month	≥ 1	Full return; stressor; postpartum specifier
Schizophreniform	1 – < 6 months	≥ 2 Criterion A	Dysfunction not required
Schizophrenia	≥ 6 months (≥ 1 month active)	≥ 2 (≥ 1 core)	Functional decline
Schizoaffective	Psychosis ≥ 2 weeks without mood	Criterion A + major mood episode	Mood for most of the illness; women
Delusional disorder	≥ 1 month	Delusion(s) only	Non-bizarre, encapsulated, function preserved
Mood disorder with psychotic features	Psychosis only **within** mood episodes	—	Mood-congruent usually

Table B: Delirium vs Schizophrenia

	Delirium	Schizophrenia
Onset	**Hours–days**	Insidious (prodrome months–years)
Consciousness	**Clouded, fluctuating, worse at night**	**Clear**
Orientation	**Impaired**	Intact
Attention	**Impaired**	Relatively intact
Hallucinations	**Visual**	**Auditory**
Age	Often elderly; any	Teens–30s
Cause	Medical/drug: find it	Primary

Table C: Antipsychotic Emergencies with Fever

	NMS	Agranulocytosis (clozapine)	Myocarditis (clozapine)	Malignant catatonia
Key	**Rigidity, ↑ CK**, autonomic instability, confusion	**↓ Neutrophils**, infection	**Weeks 4–8**, chest pain, tachycardia, **↑ troponin, eosinophils**	Catatonic signs + fever + autonomics
Test	CK, FBC	**FBC/ANC**	**ECG, troponin, CRP**	Clinical; lorazepam challenge
Action	**Stop antipsychotic**, supportive (*[added]* dantrolene/bromocriptine)	Stop clozapine; treat infection	Stop clozapine; cardiology	**Lorazepam, ECT**

Table D: Antipsychotic Side-Effect Champions

Side-effect	Worst	Spares
EPS / TD	**Haloperidol** (FGAs)	Clozapine, quetiapine *[added]*
Metabolic	**Olanzapine, clozapine** (FGA exception: chlorpromazine)	Aripiprazole *[added]*
Prolactin	**Risperidone, paliperidone**, FGAs	**Aripiprazole**
QT	**Haloperidol IV, ziprasidone, pimozide**	—
Agranulocytosis, myocarditis, ileus, seizures	**Clozapine**	—

Table E: Positive vs Negative vs Cognitive

	Positive	Negative	Cognitive
Content	Delusions, hallucinations, disorganisation, catatonia	5 A's	Attention, working memory, SOAP
Pathway	**Mesolimbic ↑ DA**	**Mesocortical ↓ DA**	Mesocortical; NMDA
Drug response	**Good (2–4 weeks)**	Poor	Poor (remediation helps)
Predicts	Presentation	**Long-term disability**	**Employment, independence**

4. Rules, Exceptions, Patterns

- **Psychosis is a symptom**; the differential starts with reversible causes.
- **Clouded consciousness → not schizophrenia. Fluctuation → delirium. Onset > 40 → organic. Non-auditory → organic.**
- **Normal hallucination-like experiences:** hypnagogic/hypnopompic, bereavement, sleep deprivation (insight, brief, no decline).
- **Delusion must be outside the patient's culture**: evil eye, jinn, sihr and ruqyah are not delusions when shared.
- **Non-bizarre delusion → delusional disorder.** Jealous subtype → violence risk; somatic subtype → dermatology/surgery.
- **Catatonia is commonest in mood disorders**, is an emergency, and responds to **lorazepam/ECT**. Malignant catatonia ≈ NMS.
- **Criterion A needs ≥ 1 of delusions, hallucinations, disorganised speech**; catatonia + negative is not enough. No single-symptom shortcut since DSM-5.
- **FRS lost special weight in DSM-5** but remain a checklist.
- **The clock:** < 1 month / 1–6 months / ≥ 6 months. **Schizoaffective = 2-week rule.**
- **Schizophrenia:** men ~21, women ~27 (+ menopausal peak); men worse.
- **Genetics:** MZ 40–50% (so environment matters); one parent 13%; both 46%.
- **Mesolimbic = therapeutic; the other three pathways = side-effects** (mesocortical worsens negative symptoms; nigrostriatal EPS; TIDA prolactin).
- **FGA ≈ SGA in efficacy, except clozapine.** Choose by side-effects.
- **Adequate trial = 4–6 weeks at a therapeutic dose; one drug; start low, go slow.**
- **Akathisia → don't increase the dose**; propranolol/reduce.
- **Fever on an antipsychotic → FBC + NMS assessment**; on clozapine also **troponin/ECG**.
- **Clozapine interruption > 48 h → re-titrate.**
- **Off medication: 60–70% relapse at 1 year, 85–90% at 2 years.** Consider an **LAI early**.
- **Physical health:** life expectancy −15–20 years; smoking is the biggest driver; **diagnostic overshadowing**.

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5. Clinical Correlations

- **Surgical/medical wards:** post-op confusion with visual hallucinations = **delirium**. Review **opioids, anticholinergics (oxybutynin), steroids**.
- **Young patient with psychosis + seizures/dyskinesia/autonomic instability → anti-NMDA receptor encephalitis**: CSF, EEG, MRI, ovarian teratoma screen *[added]*.
- **Endocrine link:** Cushing's/steroids, thyroid, Addison's, hypercalcaemia, hypopituitarism can all present with psychosis.
- **Prolactin link (Pituitary I):** antipsychotic-induced hyperprolactinaemia is a cause of amenorrhoea/galactorrhoea. Check the drug history before imaging the pituitary *[added]*.
- **ECG link:** check **QTc** before haloperidol IV; correct K^+/Mg^{2+}.
- **ED:** fever on clozapine is never "viral" until FBC, troponin and ECG are back.
- **Cardiology:** clozapine myocarditis in weeks 4–8 (↑ troponin, eosinophilia).
- **GI:** clozapine constipation → ileus. Ask about bowels at every visit.
- **Family meetings:** stress–diathesis; high expressed emotion; written early warning signs.

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6. Rapid Revision (Last-Day)

1. **Psychosis = impaired reality testing**; a symptom, not a diagnosis.
2. **Normal look-alikes:** hypnagogic/hypnopompic, bereavement, sleep deprivation.
3. **6 domains:** delusions, hallucinations, disorganised speech, disorganised behaviour, catatonia, negative.
4. **Delusion:** fixed, false, **outside culture**. Bizarre (chip) vs non-bizarre (affair → delusional disorder).
5. **Types:** persecutory (commonest) · grandiose (mania) · nihilistic (**Cotard**, elderly psychotic depression) · somatic · religious · erotomanic (**de Clérambault**, stalking) · jealous (**Othello**, alcohol, violence) · reference (earliest) · **passivity** (broadcast, insertion/withdrawal, control).
6. **Culture:** evil eye, jinn, sihr, ruqyah, healer first = **not** delusions. Ask the family.
7. **Hallucination modality:** auditory → schizophrenia; visual → delirium/Lewy/substances; olfactory → **TLE**; tactile → **cocaine/amphetamine/alcohol withdrawal**.
8. **Catatonia:** waxy flexibility, mutism, echolalia/echopraxia. Commonest in **mood disorders**; **lorazepam, ECT**; malignant ≈ NMS.
9. **Negative 5 A's:** affect, alogia, avolition, anhedonia, asociality. **Respond least**; exclude depression/EPS.
10. **FRS:** insertion, withdrawal, broadcasting, 3rd-person/commentary, made acts. ~75% of schizophrenia; **no longer special in DSM-5**.
11. **Delirium:** acute, fluctuating, clouded, disoriented, **visual**. Mortality ≈ MI.
12. **Funnel:** organic → mood → primary psychotic.
13. **Mimics: anti-NMDA**, thyroid, Cushing/Addison, Ca^{2+}, B12/thiamine/niacin, SLE, Wilson, porphyria, HIV, syphilis.
14. **Drugs:** stimulants (Captagon), cannabis, LSD/PCP, withdrawal; **steroids, anticholinergics, levodopa, isoniazid, fluoroquinolones, mefloquine, interferon, digoxin**.
15. **Red flags:** > 40, non-auditory, clouding, neuro signs, fever/autonomic, onset in hours.
16. **Work-up:** collateral + exam · CBC, U&E, LFT, Ca, glucose · TSH, B12, folate · **UDS** · HIV/syphilis · CT/MRI, EEG · **baseline weight, BMI, waist, BP, lipids, HbA1c, ECG**.
17. **Clock:** brief < 1 month (≥ 1 symptom) · schizophreniform 1–6 months · schizophrenia ≥ 6 months · delusional ≥ 1 month · schizoaffective **2-week rule**.
18. **Epidemiology:** ~1% (DSM 0.3–0.7%); men 21, women 27; life −15–20 years.
19. **Genes:** 50–70% heritability; MZ 40–50%; DZ 17%; one parent 13%; both 46%; sibling 8–10%.
20. **Dopamine:** mesolimbic ↑ (positive) · mesocortical ↓ (negative) · nigrostriatal (EPS) · TIDA (prolactin).
21. **NMDA hypofunction** (ketamine/PCP); neurodevelopmental (winter birth); **enlarged ventricles, hypofrontality**.
22. **DSM-5-TR A–F:** ≥ 2 for 1 month (≥ 1 core) · decline · ≥ 6 months · exclude mood, substance/medical, autism.
23. **Cognitive SOAP**; cognition predicts function.
24. **Prognosis:** 15 / 30 / 55. Good = female, late and sudden onset, stressor, positive/mood symptoms, good premorbid function.
25. **Suicide ~5%** (older texts 10–13%); smoking is the biggest mortality driver.
26. **FGA vs SGA:** same efficacy except **clozapine**. FGA → EPS; SGA → metabolic (olanzapine/clozapine). **Aripiprazole spares prolactin.**
27. **Trial 4–6 weeks; one drug; start low.**
28. **EPS:** dystonia hours (IM anticholinergic) → akathisia days (propranolol, ↓ dose) → parkinsonism weeks → **TD months–years** (VMAT2).
29. **NMS:** fever, rigidity, autonomic instability, confusion, ↑ CK → stop the drug.
30. **Clozapine:** ≥ 2 failed trials, suicidality, TD. **ANC**, myocarditis (weeks 4–8), seizures, **ileus**, hypersalivation. **> 48 h gap → re-titrate.**

31. **Relapse off meds:** 60–70% (1 y), 85–90% (2 y); on meds ~20%. **LAI early.** High expressed emotion → relapse.

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7. Exam-Style Questions

Q1 (lecture, slides 13–14). A 34-year-old man says his family is affected by the evil eye and requests ruqyah. He works, prays with the family and sleeps well; the whole extended family shares the belief. No hallucinations or decline. Referral: "delusional, assess for antipsychotics." Correct assessment? A. Delusion: start an antipsychotic · **B. Culturally shared belief, not a delusion** · C. Overvalued idea: CBT · D. Somatic delusion

> **Answer: B** (lecture key).
> - A delusion must be fixed, false **and** outside the cultural/religious frame.
> - Here the belief is shared, with no hallucinations, disorganisation or decline.
> - Treating him adds metabolic/EPS risk and loses the family's trust.

Q2 (lecture, slides 22–23). A 72-year-old is 3 days post hemiarthroplasty, pulling at his cannula, seeing children in the room, disoriented. He was oriented this morning and drowsy last night; on tramadol and oxybutynin; T 37.9 °C. Most likely? A. Late-onset schizophrenia · **B. Delirium: find the cause** · C. Brief psychotic disorder · D. Delusional disorder

> **Answer: B** (lecture key).
> - Acute, fluctuating, impaired consciousness/attention, disoriented, visual hallucinations.
> - Plus anticholinergic burden and fever.
> - Examine, screen infection/hypoxia, glucose, electrolytes, Ca, renal function, review drugs.

Q3 (lecture, slides 35–36). A 29-year-old woman has had persecutory delusions and voices for 5 months; for the first 3 she was profoundly depressed. Depression has settled; for 6 weeks her mood is reasonable but the voices and persecutory belief continue. Diagnosis? A. Depression with psychotic features · B. Schizophrenia · **C. Schizoaffective disorder, depressive type** · D. Brief psychotic disorder

> **Answer: C** (lecture key).
> - Psychosis has persisted **> 2 weeks without a mood episode**, and mood symptoms were present for most of the illness.
> - A fails: psychosis must occur **only** within the mood episode.
> - B fails: a full depressive episode for most of the illness is atypical.
> - D fails on duration.

Q4 (lecture, slides 45–46). A 24-year-old woman on haloperidol for 3 weeks: voices almost gone, now amenorrhoea and bilateral galactorrhoea; pregnancy test negative; prolactin very high. Pathway? A. Mesolimbic · B. Mesocortical · C. Nigrostriatal · **D. Tuberoinfundibular**

> **Answer: D** (lecture key).
> - Dopamine tonically inhibits prolactin; D2 blockade raises it.
> - Reduce the dose or switch to **aripiprazole**.

Q5 (lecture, slides 69–70). A 31-year-old, 5 weeks on clozapine, has 2 days of fever 38.6 °C, malaise, mild chest discomfort, HR 118, clear chest. Triage: "viral; discharge." Before he leaves? A. Reassure and discharge · **B. FBC, ECG, troponin, CRP; call psychiatry** · C. Increase clozapine · D. Oral antibiotics, review in a week

> **Answer: B** (lecture key).
>
> - Three clozapine emergencies present with fever: **agranulocytosis** (FBC), **myocarditis** (weeks 2–8; troponin, CRP, eosinophils), **NMS** (CK, rigidity).
> - Discharging him = diagnostic overshadowing.

Lecture board one-liners (slide 75):

#	Question	Answer
1	Psychotic symptoms 3 weeks, full return to baseline?	**Brief psychotic disorder** (1 day – < 1 month)
2	Criterion A symptoms that can't satisfy it alone?	**Catatonic behaviour and negative symptoms**
3	First episode with olfactory hallucinations: next step?	**Investigate organic cause** (TLE)
4	Pathway mediating EPS?	**Nigrostriatal**
5	Restless, pacing 2 weeks after risperidone?	**Akathisia**: reduce dose / propranolol; **don't increase**
6	When is clozapine indicated?	**≥ 2 failed adequate trials**; persistent suicidality; severe TD

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8. Exam Pitfalls & Traps

Trap	Correct thinking
"Psychotic" = schizophrenia	❌ Psychosis is a **symptom**; work down the funnel
Post-op confused patient seeing things → psychiatry for "psychosis"	❌ **Delirium**: find the medical cause
Visual/olfactory hallucinations → schizophrenia	❌ **Organic until proven otherwise** (delirium, TLE, substances)
New psychosis at 55 → late-onset schizophrenia	❌ > 40 = organic first
Shared belief in evil eye/jinn = delusion	❌ Not if the patient's community shares it
One bizarre delusion satisfies Criterion A	❌ DSM-IV only. DSM-5-TR always needs **≥ 2**
Catatonia + negative symptoms = Criterion A	❌ Need ≥ 1 of delusions/hallucinations/disorganised speech
FRS are diagnostic of schizophrenia	❌ Also in mania, psychotic depression, organic psychosis; no special DSM-5 weight
Catatonia = schizophrenia subtype	❌ Commonest in **mood disorders**; no longer a subtype; emergency; lorazepam/ECT
Psychosis persisting after depression resolves = psychotic depression	❌ ≥ 2 weeks without a mood episode → **schizoaffective**
Schizophreniform requires functional decline	❌ Dysfunction **not required**
SGAs are more effective than FGAs	❌ Similar, **except clozapine**; choose by side-effects
Restless patient on antipsychotic → increase the dose	❌ **Akathisia**: reduce the dose / propranolol

Trap	Correct thinking
Flat, slow patient on haloperidol = negative symptoms	❌ Exclude **drug-induced parkinsonism** first (reversible)
Fever on clozapine = viral	❌ FBC (agranulocytosis), troponin/ECG (myocarditis), CK (NMS)
Restart clozapine at the usual dose after 3 days missed	❌ > 48 h gap → **re-titrate** from a low dose
Imaging confirms schizophrenia	❌ Imaging **excludes** other causes; findings are not diagnostic
Suicide is the main cause of the mortality gap	❌ **Cardiovascular/metabolic disease and smoking**
Suicide risk is 10–13%	⚠️ Older figure; current estimate **~5%** (Palmer 2005). Know both
Negative symptoms improve on antipsychotics	❌ They respond **least**; psychosocial and functional interventions matter
LAI only as a last resort	❌ Consider **early**, especially after a non-adherence relapse
Historical subtypes guide treatment	❌ **Abolished** in DSM-5; know them for exams only